

# Decision 4 – Rejected Transaction Defaults:

## Stata comparison

*Verification of the COOPECON app (branch migration, HEAD b250f0c; the Decision 4 model, config and results are unchanged since d202e00 except the 60 default) against the professor's Decision 4 design document "Decision 4 - Rejected Transaction Defaults 040926-2.docx" and the Stata file Stata\_File\_Decision4\_050926.dta (280 participants). Prepared 2026-10-06 by the migration controller.*

### How to read this document

Each step shows, side by side, the Stata figure (or Stata output) and the corresponding app screen or value, states the exact settings that produce the pair, and ends with a verdict line. The Stata figures are the ones embedded in the professor's document (with all tracked changes accepted; the previous revision's figures that the tracked changes deleted are not used). Stata itself could not be run (the StataNow licence on this machine expired on 30 Aug 2026), so every Stata summary number quoted here (Obs / Mean / SD / Min / Max, counts) was computed from the columns stored in the .dta with pandas, exactly as "summarize" / "tab" would report them. App charts are exported from the exact Plotly figure objects of the running app (Chrome, Plotly.toImage); app screens are Chrome captures; app numbers come from the same seam/engine path the "Run" buttons execute.

### Settings used for every comparison (Stata parameter values)

| Setting                                                           | Value                                                                                      | Why                                                                                                                                                                                           |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Page 1 › Population Mode                                          | Research Baseline                                                                          | the original 280 participants, no stochastic component (Stata: the 280 observations)                                                                                                          |
| Page 1 › Number of Agents                                         | 280                                                                                        | $N \leq 280 \rightarrow$ the first 280 original rows in order, no random draw; seed irrelevant (42 used)                                                                                      |
| Page 1 › Income                                                   | professor's income file (research modes)                                                   | <code>z_net_income = std(income)</code> over the 280, as in the .dta                                                                                                                          |
| Page 2 › Decision 4 › Income Specification                        | Continuous only / Categorical only                                                         | WTP and Risk-Taking are the only income-using elements                                                                                                                                        |
| Page 2 › Decision 4 › Intercepts $\beta_0 \dots \beta_4$          | all 0.0000 (= app defaults)                                                                | the .dta embeds no intercept; since HEAD ffa32fb the app default $\beta_0$ (Options List Length, applied on the raw composite) is 0 as well, so the Stata comparison runs at the app defaults |
| Page 2 › Decision 4 › Flexibility Anchor Mix <code>W_OFlex</code> | 0.25 (default)                                                                             | <code>anchored_flexibility = 0.25 * z_stdactions + 0.75 * z_Flexibility_calculated_ivw</code>                                                                                                 |
| Page 2 › Decision 4 › Stochastic                                  | off (Research Baseline is always deterministic)                                            | $\sigma$ tables compared separately (step 10); a Research Specification run with $\sigma$ on is shown in step 12                                                                              |
| Histogram bins                                                    | Stata default rule $\min(\sqrt{N}, 10 \cdot \log_{10} N) \rightarrow$ 17 bins at $N = 280$ | the app applies the same rule (capped at 17); Stata plots density, the app plots the proportion of agents – identical shapes, y-axis scaled by the bin width                                  |

### Summary of verdicts

| Step | Stata item                                                     | App item                                      | Verdict                      |
|------|----------------------------------------------------------------|-----------------------------------------------|------------------------------|
| 1    | Standardisation (egen std of the Big-5, education, stdactions) | YAML <code>z_scoring</code> vs .dta summarize | Match (7 significant digits) |

|    |                                                                                                            |                                                                 |                                                                                                                                                                                                               |
|----|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2  | Element equations and coefficients (TTP, Loyalty, WTP, Risk-Taking, Flexibility)                           | doc equations vs app sub-tabs                                   | Match                                                                                                                                                                                                         |
| 3  | F1 histogram weighted_ttp, F2 choice_length_deterministic                                                  | app "Tendency to Plan score" / "Number of pre-selected options" | Match – 280/280 agents ( $\beta_0 = 0$ )                                                                                                                                                                      |
| 4  | F3 histogram weighted_loyalty, F4 weighted_loyalty15                                                       | app "Loyalty score" / loyalty-based options ranking             | Match – 280/280 (score, segment, 5-option list)                                                                                                                                                               |
| 5  | F6 histogram z_WTP_calculated, F8 WTP_calculated15 (continuous)                                            | app WTP score / ranking, Continuous only                        | Match – 280/280                                                                                                                                                                                               |
| 6  | T1 regress WTP (categorical), F5/F7 categorical WTP histograms                                             | app quintile-effects table / WTP score, Categorical only        | Coefficients match; the .dta's stored weighted_WTP_categorical was generated with a superseded formula (see step 6) – the app reproduces the document's printed code; 1 agent's segment differs from the .dta |
| 7  | F9 histogram z_RT_calculated_hs, F12 RT_calculated15                                                       | app Risk-Taking score / ranking, Continuous only                | Match – 280/280                                                                                                                                                                                               |
| 8  | T2 regress RT (categorical), F10/F11 categorical RT histograms                                             | app quintile-effects table / RT score, Categorical only         | Match – 280/280 (z-score and floor-rule segments)                                                                                                                                                             |
| 9  | F13 z_Flexibility_calculated_ivw, F14 z_stdactions, F15 z_anchored_flexibility, F16 Flexibility_combined15 | app Flexibility score / ranking + export columns                | Match – 280/280                                                                                                                                                                                               |
| 10 | $\sigma$ tables (TTP, Loyalty, WTP, Risk-Taking, Flexibility)                                              | app "Base $\sigma$ and effective $\sigma$ per budget level"     | Match the document literally (two arithmetic slips inside the document itself are noted)                                                                                                                      |
| 11 | Section 6 integrated default list (Kemeny-Young + tie-break)                                               | app "Integrated Default List" + tie statistics                  | App-only (no Stata counterpart); output rules hold 280/280                                                                                                                                                    |
| 12 | Stochastic component (no Stata figure)                                                                     | Research Specification run with $\sigma$ on                     | App-only example; UI = engine (identical allocation)                                                                                                                                                          |
| 13 | Compare both / Compare all layouts                                                                         | app comparison screens                                          | Consistent with the single-mode runs; Copula differs by construction                                                                                                                                          |

## Step 1: Settings in the app

Page 1 (population mode and agents) and the Decision 4 tab (income specification, stochastic component, intercept override):

# COOPECON AI Agent Simulation

## Page 1: Common Simulation Parameters

### Simulation Execution Mode

Execution Mode ⓘ  
☒ Snapshot ☐ Live Simulation

### Simulation Mode

Analysis Mode ⓘ  
☒ Single Run ☐ Monte-Carlo Study

Single Run: Execute one simulation with current parameters

### Simulation Settings

Number of Agents ⓘ  
280 - +  
Random Seed ⓘ

### Time Parameters

### Market Parameters

Platform Markup (m) ⓘ  
0.10  
Price Range (r) ⓘ  
0.25  
Bidding Percentage (bp) ⓘ  
0.50  
Products for auction per vendor: 50  
Price Grid Categories (g) ⓘ  
11 - +

### Annual Income Distribution

Research Baseline and Research Specification use each original participant's income from the professor's data file; these distribution settings apply to Copula populations.

### Income Categories

Categories determine customer status (discount/fixed) and purchasing limits

Discount Income Categories (NDIC) ⓘ  
10 - +  
Fixed Income Categories (NFIC) ⓘ  
10 - +

### Purchasing Limits

Limits by Fixed Income Categories (Cat 1 = lowest income, applies to discount customers)

Apply Purchasing Limits? ⓘ  
☐ Yes ☒ No

Purchasing limits are disabled. For the practical purpose of running the simulation, an artificial purchasing limit is specified here, indicating a maximum number of purchased items for all agents based on their assumed purchasing preferences and budget limit.

Note: This artificial limit represents a market-based constraint derived from customer budget preferences and needs, rather than a regulatory purchasing restriction.

Max Purchases per Term (All Agents) ⓘ  
50 - +

### Population Generation Mode

This setting applies to all decisions and determines how agents are generated

Population Mode ⓘ  
☐ Copula (synthetic) ☐ Research Specification ☒ Research Baseline ☐ Compare all

Research Baseline: Uses original 280 participants with NO stochastic component. Returns pure anchor values (deterministic).

Reset all simulation parameters (vendors, prices, distributions) to system defaults.

Reset to Default Values

# COOPECON AI Agent Simulation

## Page 2: Decision-Specific Parameters

### Decision Selection

☐ Select All Decisions

Select Decisions to Run

4. Rejected Trans... ✕

Overview 4. Rejected Transaction Defaults

### Rejected Transaction Defaults Configuration

#### Income Specification

Income Specification for Rejected Transaction Model ⓘ

- ☐ Categorical only  
☒ Continuous only  
☐ Compare both

#### Stochastic Component

Research Baseline always uses deterministic scores. Configure stochastic settings for Copula / Research Specification runs.

Copula Mode:

☐ Add Normal(anchor, σ) draw to Copula runs ⓘ

Research Specification Mode:

☒ Use Normal(anchor, σ) draw in Research Specification mode ⓘ

Options List Length sub-tab: intercept override 0 (Research Default 0.0000, the app default):

### Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates each agent's **Tendency to Plan** based on Big-5 personality traits and education, and then converts it into the number of pre-selected default options (0-5).

$$TendencyToPlan_i (TTP_i) = \beta_0 - 0.015255656 \times z_{Extraversion_i} + 0.01776386 \times z_{Agreeableness_i} + 0.01959 \times z_{Neuroticism_i} + 0.00901465 \times z_{Conscientiousness_i} + 0.0297 \times Edu_i$$

$$OptionsListLength05_i = \left\lfloor (6 - 0.0001) \times \frac{TTP_i - \min(TTP)}{\max(TTP) - \min(TTP)} \right\rfloor \in \{0, \dots, 5\}$$

Stochastic Component:

If stochastic enabled:  $TTP_i \sim \text{Normal}(\mu = \text{anchor}, \sigma)$  where the anchor is the continuous  $TTP_i$  score and  $\sigma = \text{base } \sigma \times \text{coefficient}$  (overall or per budget level); the drawn values are re-scaled over the population and re-binned into the 0-5 options list length.

#### Intercept Override

Research Default ( $\beta_0$ ): 0.0000

#### Override Value

Baseline Options List Length tendency ⓘ

0,0000

Change

## Step 2: The standardisation of values

Stata commands (document, section 5 and the per-element "Variable Standardisation" blocks):

```
egen z_extraversionbig5 = std(extraversionbig5)
egen z_agreeable = std(agreeable)
egen z_neuroticismbig5 = std(neuroticismbig5)
egen z_conscientiousnessbig5 = std(conscientiousnessbig5)
egen z_opennessbig5 = std(opennessbig5)
egen z_net_income = std(income)
gen reduction = education-1
egen z_stdactions = std(stdactions)
```

The app freezes the original-280 means and SDs in config/decisions.yaml (rejected\_transaction\_defaults.z\_scoring) and uses the professor's income file for the research modes. "summarize" of the raw .dta columns vs the app constants:

| Variable                 | Obs | Stata mean | Stata SD  | App mean                                                   | App SD    | Equal                                                            |
|--------------------------|-----|------------|-----------|------------------------------------------------------------|-----------|------------------------------------------------------------------|
| extraversionbig5         | 280 | 3.557857   | 0.6989565 | 3.557857                                                   | 0.6989565 | yes                                                              |
| agreeable                | 280 | 3.546071   | 0.3732712 | 3.546071                                                   | 0.3732712 | yes                                                              |
| neuroticismbig5          | 280 | 2.702143   | 0.6839657 | 2.702143                                                   | 0.6839657 | yes                                                              |
| conscientiousnessbig5    | 280 | 3.657143   | 0.5596521 | 3.657143                                                   | 0.5596521 | yes                                                              |
| opennessbig5             | 280 | 4.060714   | 0.5068274 | 4.060714                                                   | 0.5068274 | yes                                                              |
| stdactions               | 280 | 1.186338   | 0.6752843 | 1.186338                                                   | 0.6752843 | yes                                                              |
| income<br>(z_net_income) | 280 | 28334.95   | 17711.6   | computed per<br>run from the<br>professor's<br>income file |           | yes (engine<br>income stats<br>mean<br>28334.95, sd<br>17711.60) |
| reduction                | 280 | 0.3785714  | 0.4858995 | Education – 1<br>per agent                                 |           | yes                                                              |

The values match exactly (to the 7 significant digits Stata prints).

## Step 2b: The element equations

The document's final equations and the app's sub-tab formulas, coefficient by coefficient:

| Element                               | Document                                                                                                                                                                          | App (config/decisions.yaml,<br>shown on the sub-tabs)                                                    | Equal |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------|
| TTP (doc §1)                          | $\beta_0 - 0.0152556564 \cdot zE + 0.0177638642 \cdot zA + 0.01959 \cdot zN + 0.00901465 \cdot zC + 0.0297 \cdot reduction$                                                       | $-0.0152556564 / 0.0177638642 / 0.01959 / 0.00901465 / 0.0297$                                           | yes   |
| Loyalty (doc §2, bs_weighted_loyalty) | $\beta_1 - 0.009828468 \cdot zE + 0.01096706 \cdot zO + 0.0123046 \cdot zA$ (then egen std)                                                                                       | $-0.009828468 / 0.01096706 / 0.0123046$                                                                  | yes   |
| WTP (doc §3, H&S)                     | $\beta_2 + 0.078863062 \cdot zE - 0.012326128 \cdot zA + 0.698 \cdot z\_net\_income$                                                                                              | $0.078863062 / -0.012326128 / 0.698$                                                                     | yes   |
| Risk-Taking (doc §4, H&S)             | $\beta_3 + 0.025942386297 \cdot zE + 0.023699214948 \cdot zO - 0.038734315188 \cdot zA - 0.037739440732 \cdot zC - 0.025388697852 \cdot zN + 0.006874197106 \cdot z\_net\_income$ | $0.025942386297 / 0.023699214948 / -0.038734315188 / -0.037739440732 / -0.025388697852 / 0.006874197106$ | yes   |
| Flexibility (doc §5, IVW)             | $\beta_4 + 0.0206 \cdot zE + 0.0294118 \cdot zO - 0.04921357 \cdot zN + 0.04339814 \cdot zA + 0.04811179 \cdot zC$ ; anchored = $0.25 \cdot z\_stdactions + 0.75 \cdot z\_calc$   | $0.0206 / 0.0294118 / -0.04921357 / 0.04339814 / 0.04811179$ ; W_OFlex 0.25                              | yes   |
| Priority sequences                    | Loyalty 3>1>4>5>2; WTP 3>2>1>4>5; RT 4>2>1>3>5; Flexibility 2>4>3>1>5                                                                                                             | loyalty 3>1>4>5>2; wtp 3>2>1>4>5; risk_taking 4>2>1>3>5; flexibility 2>4>3>1>5                           | yes   |

## Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. **Loyalty Ranking** 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates *Loyalty to the vendor* based on Big-5 personality traits.

$$Loyalty_i = \beta_1 - 0.009828468 \times z_{Extroversion_i} + 0.01096706 \times z_{Openness_i} + 0.0123046 \times z_{Agreeableness_i}$$

$$Loyalty_{15_i} = \left\lceil 1 + (5 - 0.0001) \times \frac{Loyalty_i - \min(Loyalty)}{\max(Loyalty) - \min(Loyalty)} \right\rceil \in \{1, \dots, 5\}$$

Loyalty rejected transaction options sequence: Option 3 > Option 1 > Option 4 > Option 5 > Option 2

| Loyalty score segment                    | Priority list for rejected transaction options | Options list length |
|------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Loyalty score segment)  | 2                                              | 1                   |
| 2                                        | 5 > 2                                          | 2                   |
| 3                                        | 4 > 5 > 2                                      | 3                   |
| 4                                        | 1 > 4 > 5 > 2                                  | 4                   |
| 5 (Highest 20% of Loyalty score segment) | 3 > 1 > 4 > 5 > 2                              | 5                   |

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Stochastic Component:

If stochastic enabled:  $Loyalty_{15_i} \sim \text{Normal}(\mu = \text{anchor}, \sigma)$  where the anchor is the continuous  $Loyalty_i$  score and  $\sigma = \text{base } \sigma \times \text{coefficient}$  (overall or per budget level); the drawn values are re-scaled over the available range based into the 1-5 scale segment

## Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. **Willingness-to-Pay Ranking** 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates *Willingness to Pay* based on personality traits and income.

$$WTP_i = \beta_2 + 0.078863062 \times z_{Extroversion_i} - 0.012326128 \times z_{Agreeableness_i} + 0.698 \times z_{Income_i}$$

$$WTP_{15_i} = \left\lceil 1 + (5 - 0.0001) \times \frac{WTP_i - \min(WTP)}{\max(WTP) - \min(WTP)} \right\rceil \in \{1, \dots, 5\}$$

Willingness-to-Pay rejected transaction options sequence: Option 3 > Option 2 > Option 1 > Option 4 > Option 5

| Willingness-to-Pay score segment                    | Priority list for rejected transaction options | Options list length |
|-----------------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Willingness-to-Pay score segment)  | 5                                              | 1                   |
| 2                                                   | 4 > 5                                          | 2                   |
| 3                                                   | 1 > 4 > 5                                      | 3                   |
| 4                                                   | 2 > 1 > 4 > 5                                  | 4                   |
| 5 (Highest 20% of Willingness-to-Pay score segment) | 3 > 2 > 1 > 4 > 5                              | 5                   |

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Stochastic Component:

## Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Risk-Taking propensity based on Big-5 personality traits and income.

$$RiskTaking_i = \beta_3 + 0.0259423863 \times z_{Extroversion_i} + 0.0236992149 \times z_{Openness_i} - 0.0387343152 \times z_{Agreeableness_i} - 0.0377394407 \times z_{Conscientiousness_i} - 0.0253886979 \times z_{Neuroticism_i}$$

$$RiskTaking15_i = \left\lfloor 1 + (5 - 0.0001) \times \frac{RiskTaking_i - \min(RiskTaking)}{\max(RiskTaking) - \min(RiskTaking)} \right\rfloor \in \{1, \dots, 5\}$$

Risk-Taking rejected transaction options sequence: Option 4 > Option 2 > Option 1 > Option 3 > Option 5

| Risk-Taking score segment                    | Priority list for rejected transaction options | Options list length |
|----------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Risk-Taking score segment)  | 5                                              | 1                   |
| 2                                            | 3 > 5                                          | 2                   |
| 3                                            | 1 > 3 > 5                                      | 3                   |
| 4                                            | 2 > 1 > 3 > 5                                  | 4                   |
| 5 (Highest 20% of Risk-Taking score segment) | 4 > 2 > 1 > 3 > 5                              | 5                   |

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

Option 3: current vendor at PN price

Option 4: place a bid

Option 5: forgo the transaction

Research Baseline

## Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Flexibility based on Big-5 personality traits and observed flexibility in the experiment (captured by standard deviation in the number of actions across the eight cycle-weeks).

$$CalculatedFlexibility_i = \beta_4 + 0.0206 \times z_{Extroversion_i} + 0.0294118 \times z_{Openness_i} - 0.04921357 \times z_{Neuroticism_i} + 0.04339814 \times z_{Agreeableness_i} + 0.04811179 \times z_{Conscientiousness_i}$$

$$AnchoredFlexibility_i = W_{Flex} \times ObservedFlexibility_i + W_{Flex} \times CalculatedFlexibility_i = 0.25 \times ObservedFlexibility_i + 0.75 \times CalculatedFlexibility_i$$

ObservedFlexibility is the observed flexibility variable stdactions (the standard deviation in the number of actions across the eight cycle-weeks): a copula trait for synthetic populations and the participant's own value in the research baseline and research specification modes. All variables are standardized prior to calculation.

$$Flexibility15_i = \left\lfloor 1 + (5 - 0.0001) \times \frac{AnchoredFlexibility_i - \min(AnchoredFlexibility)}{\max(AnchoredFlexibility) - \min(AnchoredFlexibility)} \right\rfloor \in \{1, \dots, 5\}$$

Flexibility rejected transaction options sequence: Option 2 > Option 3 > Option 1 > Option 5

| Flexibility score segment                    | Priority list for rejected transaction options | Options list length |
|----------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Flexibility score segment)  | 5                                              | 1                   |
| 2                                            | 1 > 5                                          | 2                   |
| 3                                            | 3 > 1 > 5                                      | 3                   |
| 4                                            | 4 > 3 > 1 > 5                                  | 4                   |
| 5 (Highest 20% of Flexibility score segment) | 2 > 4 > 3 > 1 > 5                              | 5                   |

Option 1: higher price category, same vendor

Option 2: other vendor at lower PN price

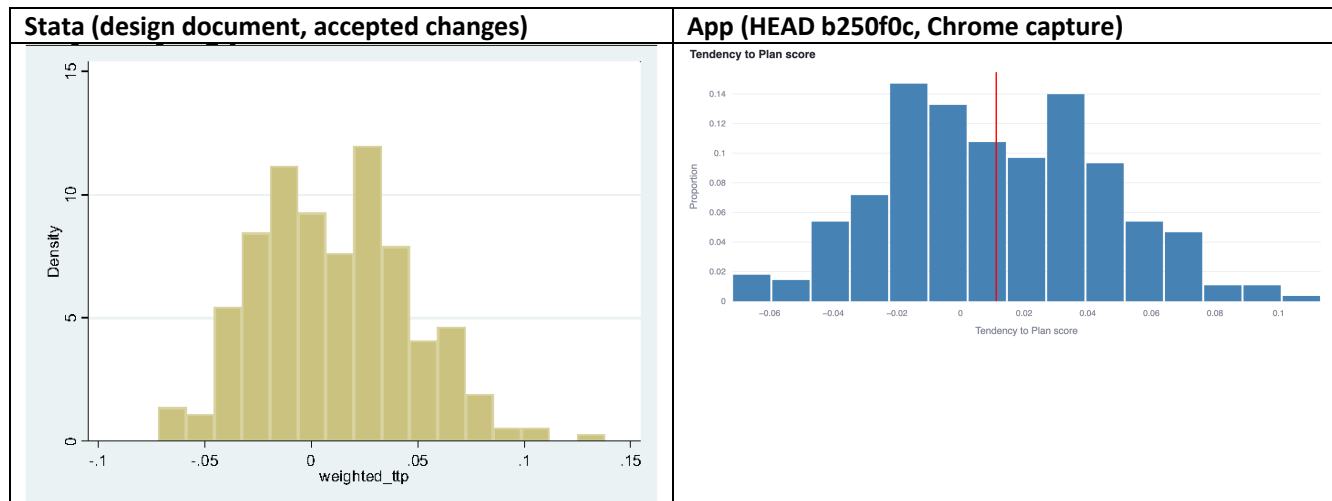
The equations and every coefficient match the document.

## Step 3: Options List Length (Tendency to Plan) – F1, F2

Settings: Research Baseline, 280 agents,  $\beta_0 = 0$  (the app default since HEAD ffa32fb; a non-zero  $\beta_0$  would shift the raw score and, with the app's fixed-cutoff semantics, the allocation; the Stata file has no intercept). Income specification irrelevant (no income term).

```
gen weighted_ttp = (-0.0152556564 * z_extraversionbig5) + (0.0177638642 * z_agreeable) + (0.01959
* z_neuroticismbig5) + (0.00901465 * z_conscientiousnessbig5) + (0.0297 * reduction)
summarize weighted_ttp
histogram weighted_ttp
```

| Variable     | Stat<br>a<br>Obs | Stata<br>Mean | Stata SD   | Stata Min   | Stata<br>Max | Ap<br>p<br>Obs | App Mean   | App SD     | App Min     | App Max   |
|--------------|------------------|---------------|------------|-------------|--------------|----------------|------------|------------|-------------|-----------|
| weighted_ttp | 280              | 0.01124357    | 0.03528479 | -0.07178839 | 0.1379994    | 280            | 0.01124358 | 0.03528479 | -0.07178839 | 0.1379994 |



```

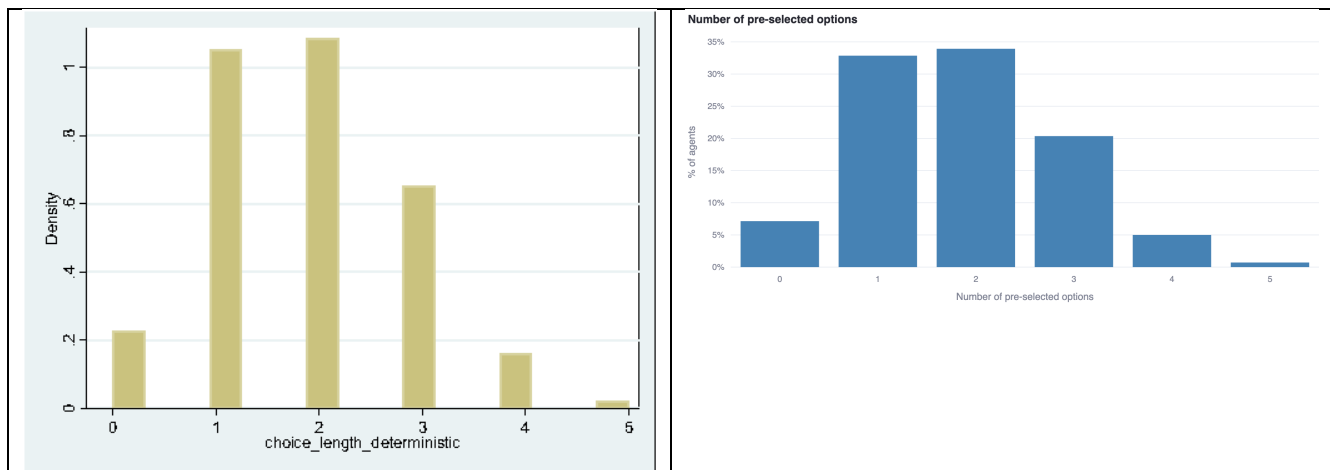
gen weighted_ttp06 = (6-0.0001)*((weighted_ttp - min_weighted_ttp)/(max_weighted_ttp - min_weighted_ttp))
gen choice_length_deterministic = floor(weighted_ttp06)

```

| Variable                    | Stata<br>Obs | Stata<br>Mean | Stata SD | Stata<br>Min | Stata<br>Max | App<br>Obs | App<br>Mean | App SD   | App<br>Min | App<br>Max |
|-----------------------------|--------------|---------------|----------|--------------|--------------|------------|-------------|----------|------------|------------|
| choice_length_deterministic | 280          | 1.853571      | 1.031748 | 0            | 5            | 280        | 1.853571    | 1.031748 | 0          | 5          |

| Category  | Stata count | Stata % | App count | App % | Equal |
|-----------|-------------|---------|-----------|-------|-------|
| options 0 | 20          | 7.1%    | 20        | 7.1%  | yes   |
| options 1 | 92          | 32.9%   | 92        | 32.9% | yes   |
| options 2 | 95          | 33.9%   | 95        | 33.9% | yes   |
| options 3 | 57          | 20.4%   | 57        | 20.4% | yes   |
| options 4 | 14          | 5.0%    | 14        | 5.0%  | yes   |
| options 5 | 2           | 0.7%    | 2         | 0.7%  | yes   |

| Stata (design document, accepted changes) | App (HEAD b250f0c, Chrome capture) |
|-------------------------------------------|------------------------------------|
|-------------------------------------------|------------------------------------|



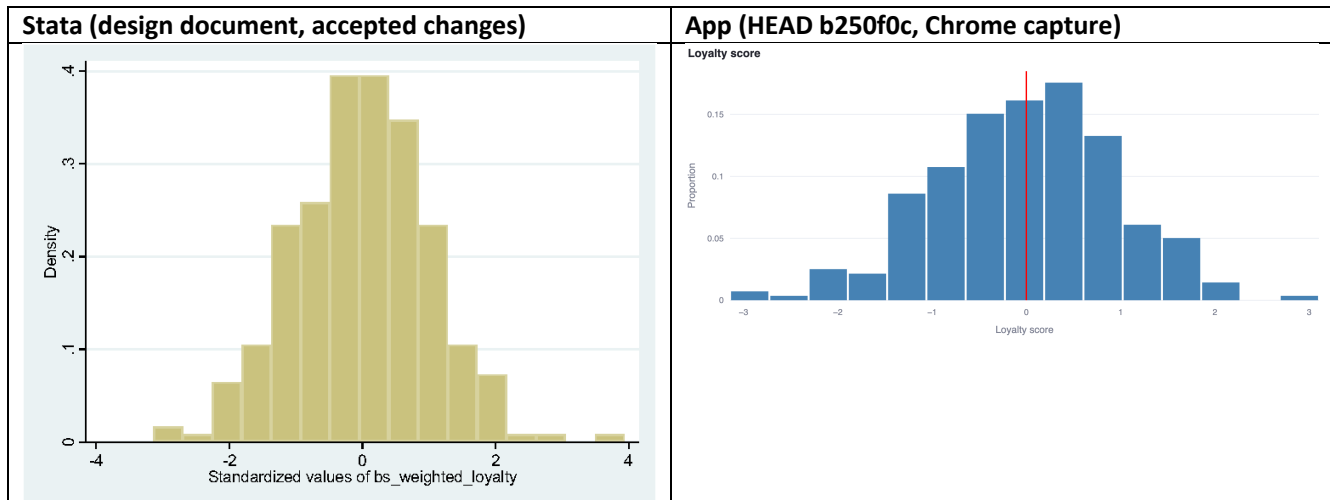
The values match exactly: `weighted_ttp` agrees for 280/280 participants (max |diff| 1.8e-08) and `choice_length_deterministic` for 280/280; the 17-bin histograms and the 0–5 allocation are identical (Stata plots density, the app the proportion).

## Step 4: Loyalty ranking – F3, F4

Settings as in step 3 ( $\beta_1 = 0$ ; no income term).

```
gen bs_weighted_loyalty = -0.009828468*z_extraversionbig5 + 0.01096706*z_opennessbig5 +
0.0123046*z_agreeable
egen weighted_loyalty = std(bs_weighted_loyalty)
replace weighted_loyalty = weighted_loyalty + beta1
summarize weighted_loyalty
histogram weighted_loyalty
```

| Variable         | Stata Obs | Stata Mean    | Stata SD | Stata Min | Stata Max | App Obs | App Mean      | App SD | App Min   | App Max |
|------------------|-----------|---------------|----------|-----------|-----------|---------|---------------|--------|-----------|---------|
| weighted_loyalty | 280       | -1.250633e-09 | 1        | -3.141984 | 3.93632   | 280     | -3.806479e-17 | 1      | -3.141984 | 3.93632 |



```

gen weighted_loyalty15 = floor(1+(5-0.0001)*((weighted_loyalty -
min_weighted_loyalty)/(max_weighted_loyalty - min_weighted_loyalty)))
summarize weighted_loyalty15
histogram weighted_loyalty15
replace choicel_loyalty_deterministic = 2 if weighted_loyalty15 == 1 ... = 3 if
weighted_loyalty15 == 5

```

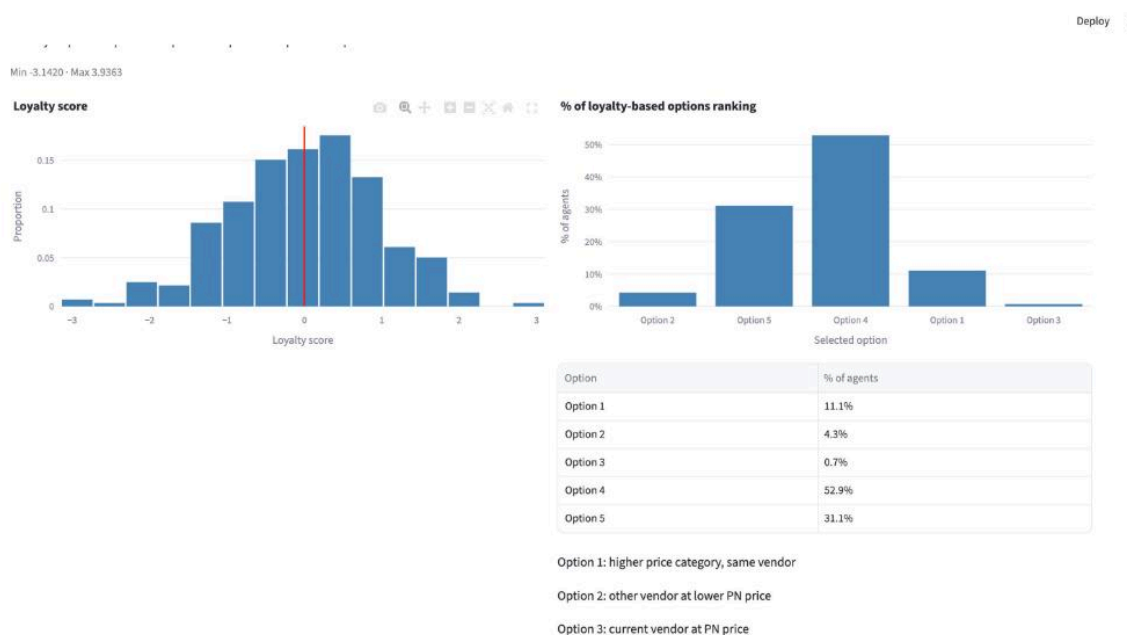
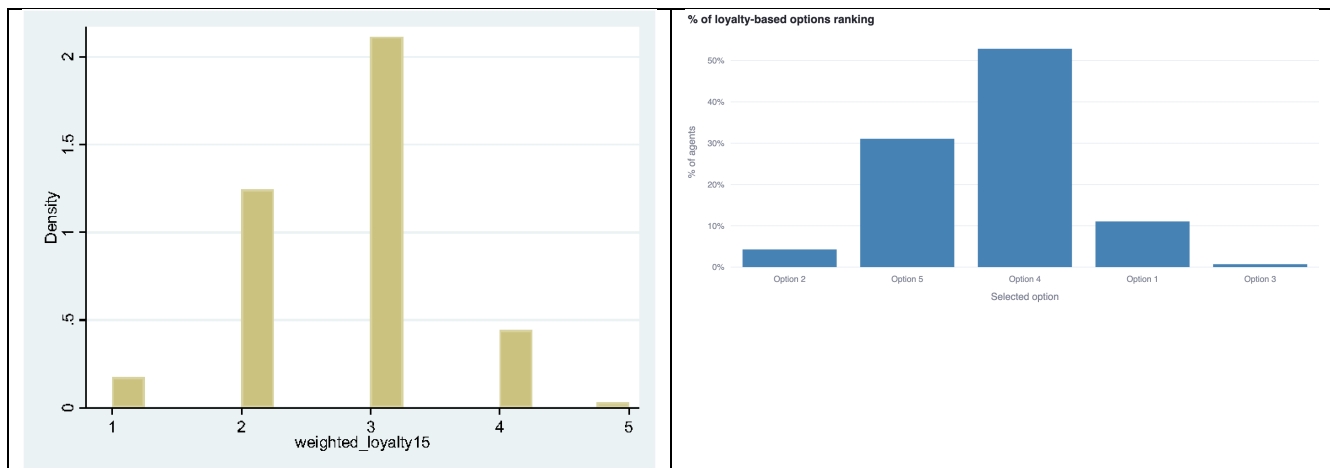
The app chart shows the FIRST option of each agent's loyalty list, with the x-axis in reversed priority order (2, 5, 4, 1, 3) – i.e. exactly segments 1...5 of weighted\_loyalty15 relabelled with their first option:

| Variable           | Stata Obs | Stata Mean | Stata SD | Stata Min | Stata Max | App Obs | App Mean | App SD  | App Min | App Max |
|--------------------|-----------|------------|----------|-----------|-----------|---------|----------|---------|---------|---------|
| weighted_loyalty15 | 280       | 2.728571   | 0.74143  | 1         | 5         | 280     | 2.728571 | 0.74143 | 1       | 5       |

| Category  | Stata count | Stata % | App count | App % | Equal |
|-----------|-------------|---------|-----------|-------|-------|
| segment 1 | 12          | 4.3%    | 12        | 4.3%  | yes   |
| segment 2 | 87          | 31.1%   | 87        | 31.1% | yes   |
| segment 3 | 148         | 52.9%   | 148       | 52.9% | yes   |
| segment 4 | 31          | 11.1%   | 31        | 11.1% | yes   |
| segment 5 | 2           | 0.7%    | 2         | 0.7%  | yes   |

| Category       | Stata count | Stata % | App count | App % | Equal |
|----------------|-------------|---------|-----------|-------|-------|
| first option 1 | 31          | 11.1%   | 31        | 11.1% | yes   |
| first option 2 | 12          | 4.3%    | 12        | 4.3%  | yes   |
| first option 3 | 2           | 0.7%    | 2         | 0.7%  | yes   |
| first option 4 | 148         | 52.9%   | 148       | 52.9% | yes   |
| first option 5 | 87          | 31.1%   | 87        | 31.1% | yes   |

| Stata (design document, accepted changes) | App (HEAD b250f0c, Chrome capture) |
|-------------------------------------------|------------------------------------|
|-------------------------------------------|------------------------------------|



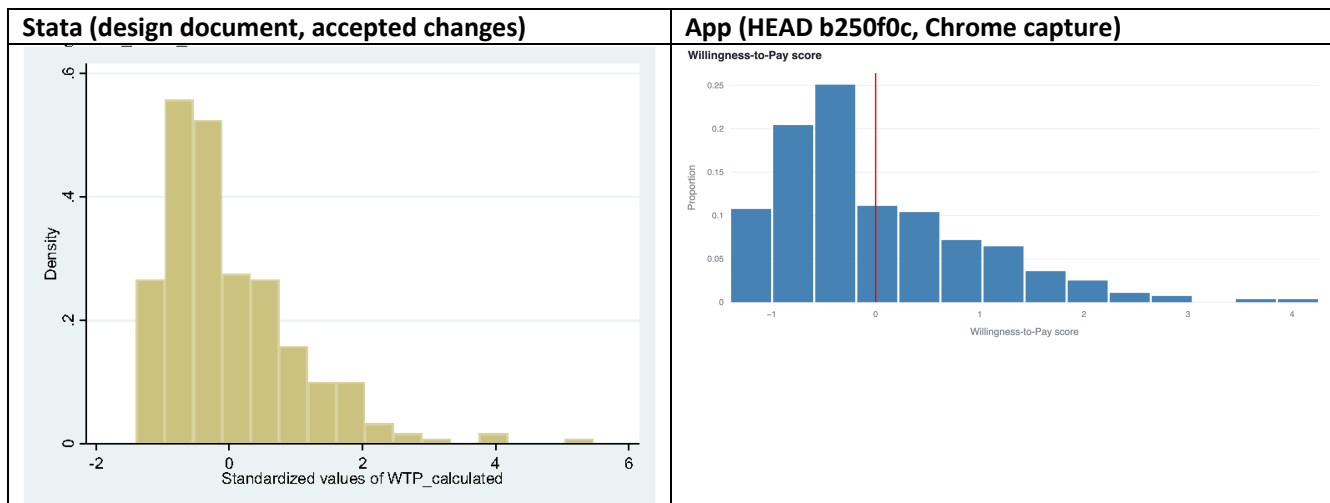
The values match exactly: weighted\_loyalty 280/280 (max |diff| 3.2e-07), weighted\_loyalty15 280/280, and the five-position choice lists choice1...choice5\_loyalty\_deterministic 280/280.

## Step 5: Willingness-to-Pay ranking, continuous income – F6, F8

Settings as in step 3 with Income Specification = Continuous only ( $\beta_2 = 0$ ).

```
gen WTP_calculated = 0.078863062 * z_extraversionbig5 - 0.012326128* z_agreeable + 0.698 *
z_net_income
egen z_WTP_calculated= std(WTP_calculated)
replace z_WTP_calculated = z_WTP_calculated + beta2
summarize z_WTP_calculated
histogram z_WTP_calculated
```

| Variable         | Stata Obs | Stata Mean   | Stata SD | Stata Min | Stata Max | App Obs | App Mean      | App SD | App Min   | App Max  |
|------------------|-----------|--------------|----------|-----------|-----------|---------|---------------|--------|-----------|----------|
| z_WTP_calculated | 280       | 1.736251e-09 | 1        | -1.397873 | 5.470183  | 280     | -1.268826e-17 | 1      | -1.397877 | 5.470187 |



```

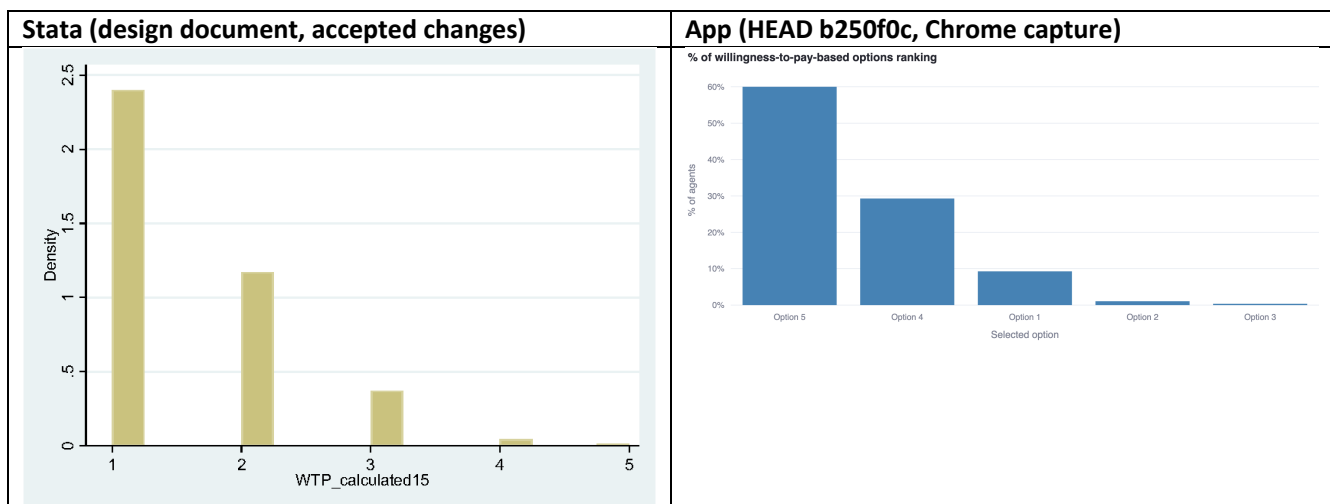
gen WTP_calculated15 = floor(1+(5-0.0001)*(( z_WTP_calculated - min_WTP_calculated)/(
max_WTP_calculated - min_WTP_calculated)))
histogram WTP_calculated15

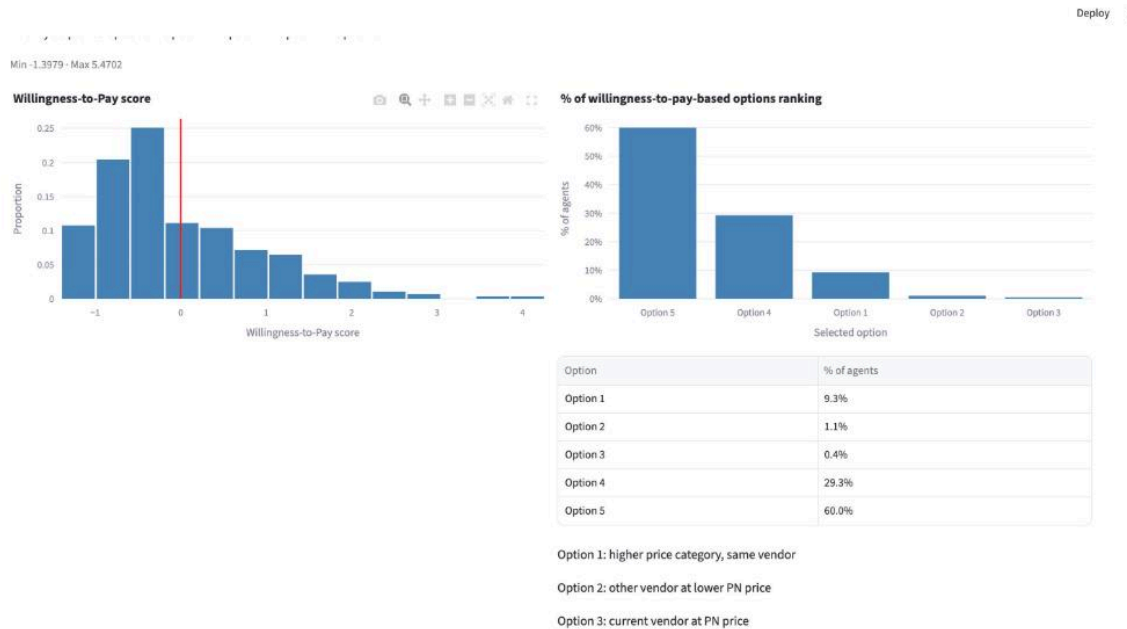
```

| Variable         | Stata Obs | Stata Mean | Stata SD  | Stata Min | Stata Max | App Obs | App Mean | App SD    | App Min | App Max |
|------------------|-----------|------------|-----------|-----------|-----------|---------|----------|-----------|---------|---------|
| WTP_calculated15 | 280       | 1.525      | 0.7376826 | 1         | 5         | 280     | 1.525    | 0.7376826 | 1       | 5       |

| Category  | Stata count | Stata % | App count | App % | Equal |
|-----------|-------------|---------|-----------|-------|-------|
| segment 1 | 168         | 60.0%   | 168       | 60.0% | yes   |
| segment 2 | 82          | 29.3%   | 82        | 29.3% | yes   |
| segment 3 | 26          | 9.3%    | 26        | 9.3%  | yes   |
| segment 4 | 3           | 1.1%    | 3         | 1.1%  | yes   |
| segment 5 | 1           | 0.4%    | 1         | 0.4%  | yes   |

| Category       | Stata count | Stata % | App count | App % | Equal |
|----------------|-------------|---------|-----------|-------|-------|
| first option 1 | 26          | 9.3%    | 26        | 9.3%  | yes   |
| first option 2 | 3           | 1.1%    | 3         | 1.1%  | yes   |
| first option 3 | 1           | 0.4%    | 1         | 0.4%  | yes   |
| first option 4 | 82          | 29.3%   | 82        | 29.3% | yes   |
| first option 5 | 168         | 60.0%   | 168       | 60.0% | yes   |





The values match exactly:  $z\_WTP\_calculated$  280/280 (max |diff| 8.1e-06, the .dta stores float32),  $WTP\_calculated15$  280/280, choice lists 280/280.

## Step 6: Willingness-to-Pay ranking, categorical income – T1, F5, F7

Settings as in step 3 with Income Specification = Categorical only ( $\beta_2 = 0$ ).

```
regress WTP_calculated WTP_noincome i.totalallowance
      WTP_noincome .5439353 | 32 .2671588 | 72 .5057716 | 128 .9411747 | 200 1.822471 | _cons -
      .691843
gen weighted_WTP_categorical = cond(assignedallowancelevel ==5, 1.822471-0.691843 + 0.078863062 *
z_extraversionbig5 - 0.012326128 * z_agreeable, cond(... ==4, 0.9411747-0.691843 + ..., cond(...
==3, 0.5057716-0.691843 + ..., cond(... ==2, 0.2671588-0.691843 + ..., -0.691843 + 0.078863062 *
z_extraversionbig5 - 0.012326128 * z_agreeable)))
egen z_WTP_categorical = std(weighted_WTP_categorical)
replace z_WTP_categorical = z_WTP_categorical+ beta2
```

The regression table's level effects appear in the app as the "Income Quintile Effects ( $\beta\_income\_q$ )" table (base intercept + level dummy):

| Quintile  | Stata (regress, document)          | App $\beta\_income\_q$ | Equal |
|-----------|------------------------------------|------------------------|-------|
| Q1 (€12)  | -.691843                           | -0.6918430             | yes   |
| Q2 (€32)  | -.691843 + .2671588 =<br>-.4246842 | -0.4246842             | yes   |
| Q3 (€72)  | -.691843 + .5057716 =<br>-.1860714 | -0.1860714             | yes   |
| Q4 (€128) | -.691843 + .9411747 =<br>.2493317  | 0.2493317              | yes   |
| Q5 (€200) | -.691843 + 1.822471 =<br>1.130628  | 1.1306280              | yes   |

## Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. **Willingness-to-Pay Ranking** 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Willingness to Pay based on personality traits and income.

$$WTP_i = \beta_2 + 0.078863062 \times z_{Extroversion_i} - 0.012326128 \times z_{Agreeableness_i} + \beta_{income\_q}[quintile_i]$$

Income Quintile Effects ( $\beta_{income\_q}$ ):

| Quintile  | $\beta_{income\_q}$ |
|-----------|---------------------|
| Q1 (€12)  | -0.6918430          |
| Q2 (€32)  | -0.4246842          |
| Q3 (€72)  | -0.1860714          |
| Q4 (€128) | 0.2493317           |
| Q5 (€200) | 1.1306280           |

$\beta_{income\_q}$ : Income quintile effects based on agent's income category (Quintiles 1-5)

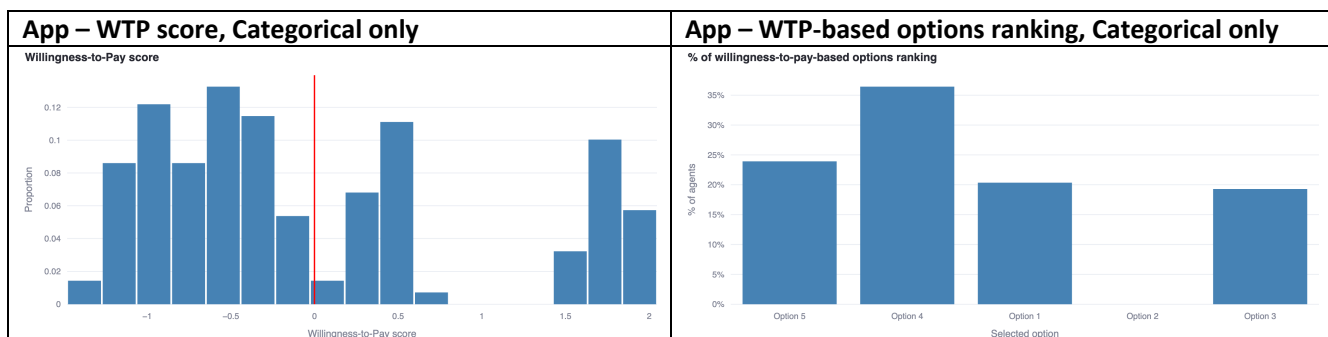
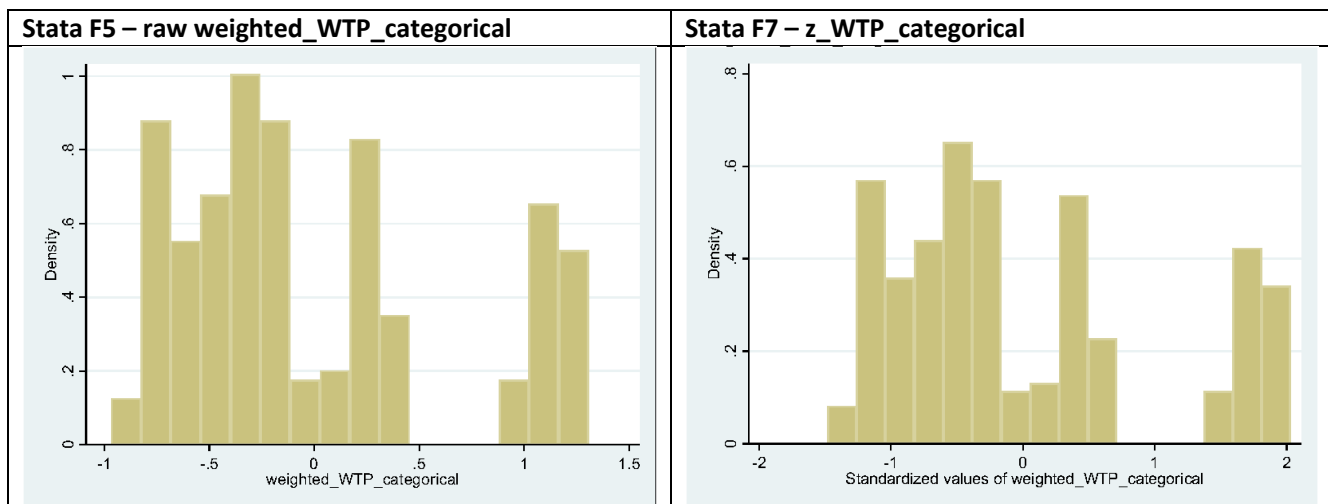
Each value = base intercept (-0.6918430) + the quintile's differential income effect (Quintile 1 is the base level).

$$WTP15_i = \left\lfloor 1 + (5 - 0.0001) \times \frac{WTP_i - \min(WTP)}{\max(WTP) - \min(WTP)} \right\rfloor \in \{1, \dots, 5\}$$

Willingness-to-Pay rejected transaction options sequence: Option 3 > Option 2 > Option 1 > Option 4 > Option 5

| Willingness-to-Pay score segment                   | Priority list for rejected transaction options | Options list length |
|----------------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Willingness-to-Pay score segment) | 5                                              | 1                   |

Stata figures F5 (raw weighted\_WTP\_categorical) and F7 (standardised) vs the app's categorical WTP score (the app charts the standardised score; F5 is the same histogram on the raw scale):





| Variable                                 | Stat<br>a<br>Obs | Stata Mean        | Stata SD      | Stata<br>Min   | Stata<br>Max | Ap<br>p<br>Obs | App Mean         | App SD        | App Min        | App<br>Max   |
|------------------------------------------|------------------|-------------------|---------------|----------------|--------------|----------------|------------------|---------------|----------------|--------------|
| z_WTP_categorical<br>(.dta)              | 280              | -2.328306e<br>-10 | 1             | -1.48075<br>2  | 2.0299<br>77 | 28<br>0        | 3.172066e-<br>18 | 1             | -1.47787<br>3  | 2.0466<br>8  |
| weighted_WTP_categ<br>orical (.dta, raw) | 280              | -0.0086953<br>3   | 0.64771<br>16 | -0.96779<br>56 | 1.3061<br>44 | 28<br>0        | -4.3748e-0<br>8  | 0.63817<br>88 | -0.94314<br>73 | 1.3061<br>48 |

Finding (document / .dta error, not an app defect). The app evaluates the document's printed `cond()` code exactly ( $\max |diff| 9e-09$  against a direct re-computation of that code on the .dta z-columns), but the column `weighted_WTP_categorical` STORED in `Stata_File_Decision4_050926.dta` was generated with a superseded version of that command: for allowance levels 2–5 it uses the agreeableness coefficient  $-0.0123242128$  (the value the tracked changes of rev 040926-2 deleted in favour of  $-0.012326128$ ), and for level 1 (57 participants) a different intercept and coefficient altogether (fitted exactly:  $-0.7355455 + 0.078863062 \cdot zE - 0.0046002694 \cdot zA$ , instead of  $-0.691843 + 0.078863062 \cdot zE - 0.012326128 \cdot zA$ ). Consequently the .dta's `z_WTP_categorical` differs from the document's own formula by up to 0.062 (z-units), its min/max are  $-1.480752 / 2.029977$  versus  $-1.477873 / 2.046680$  for the printed code, and the F7 histogram above is the histogram of the superseded column. The segment allocation differs for exactly 1 of 280 participants (Stata-stored column: 67/101/58/0/54; document code = app: 67/102/57/0/54). The app follows the written document; the professor may want to regenerate the .dta column with the current `cond()` command.

| Category                               | Stata count | Stata % | App count | App % | Equal |
|----------------------------------------|-------------|---------|-----------|-------|-------|
| segment (Stata-stored column vs app) 1 | 67          | 23.9%   | 67        | 23.9% | yes   |
| segment (Stata-stored column vs app) 2 | 101         | 36.1%   | 102       | 36.4% | NO    |
| segment (Stata-stored column vs app) 3 | 58          | 20.7%   | 57        | 20.4% | NO    |

|                                        |    |       |    |       |     |
|----------------------------------------|----|-------|----|-------|-----|
| segment (Stata-stored column vs app) 4 | 0  | 0.0%  | 0  | 0.0%  | yes |
| segment (Stata-stored column vs app) 5 | 54 | 19.3% | 54 | 19.3% | yes |

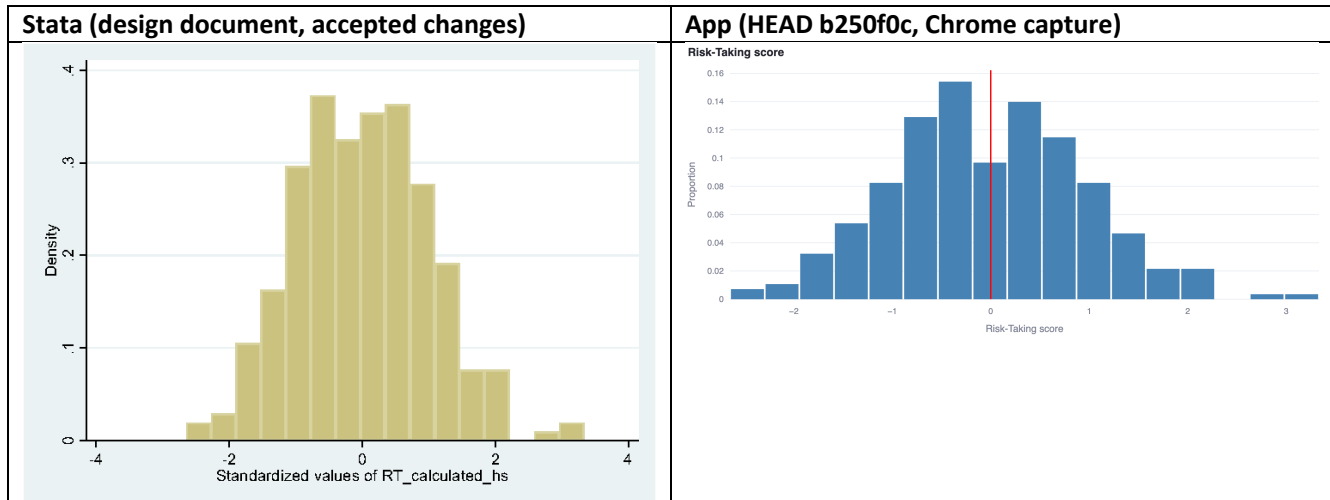
The coefficients match the document exactly; the app reproduces the document's categorical WTP code to 1e-08. The stored .dta column is the only mismatch (superseded formula in the file, 1 agent's segment).

## Step 7: Risk-Taking ranking, continuous income – F9, F12

Settings as in step 5 ( $\beta_3 = 0$ , Continuous only).

```
gen RT_calculated_hs = 0.025942386297 * z_extraversionbig5 + 0.023699214948 * z_opennessbig5 -
0.038734315188 * z_agreeable - 0.037739440732 * z_conscientiousnessbig5 - 0.025388697852 *
z_neuroticismbig5 + 0.006874197106 * z_net_income
egen z_RT_calculated_hs = std(RT_calculated_hs)
replace z_RT_calculated_hs = z_RT_calculated_hs + beta3
histogram z_RT_calculated_hs
```

| Variable               | Stat<br>a<br>Obs | Stata Mean        | Stat<br>a SD | Stata Min     | Stata Max    | Ap<br>p<br>Obs | App Mean          | Ap<br>p<br>SD | App Min       | App Max      |
|------------------------|------------------|-------------------|--------------|---------------|--------------|----------------|-------------------|---------------|---------------|--------------|
| z_RT_calculated_h<br>s | 280              | -8.421401e-1<br>0 | 1            | -2.64659<br>4 | 3.33315<br>3 | 280            | -2.854859e-1<br>7 | 1             | -2.64659<br>4 | 3.33315<br>3 |



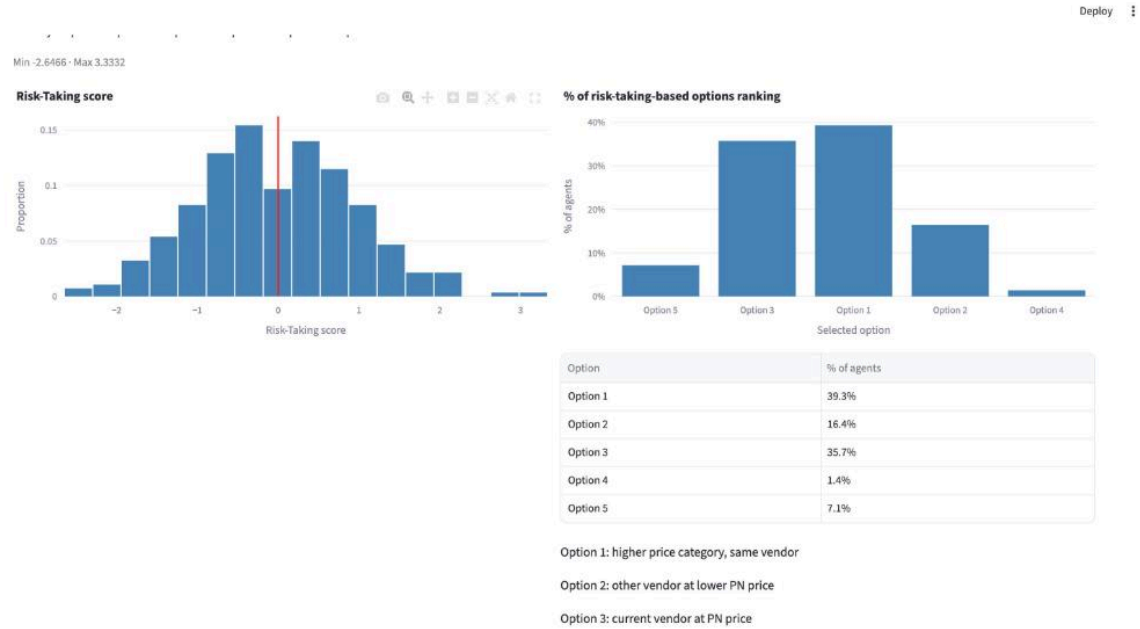
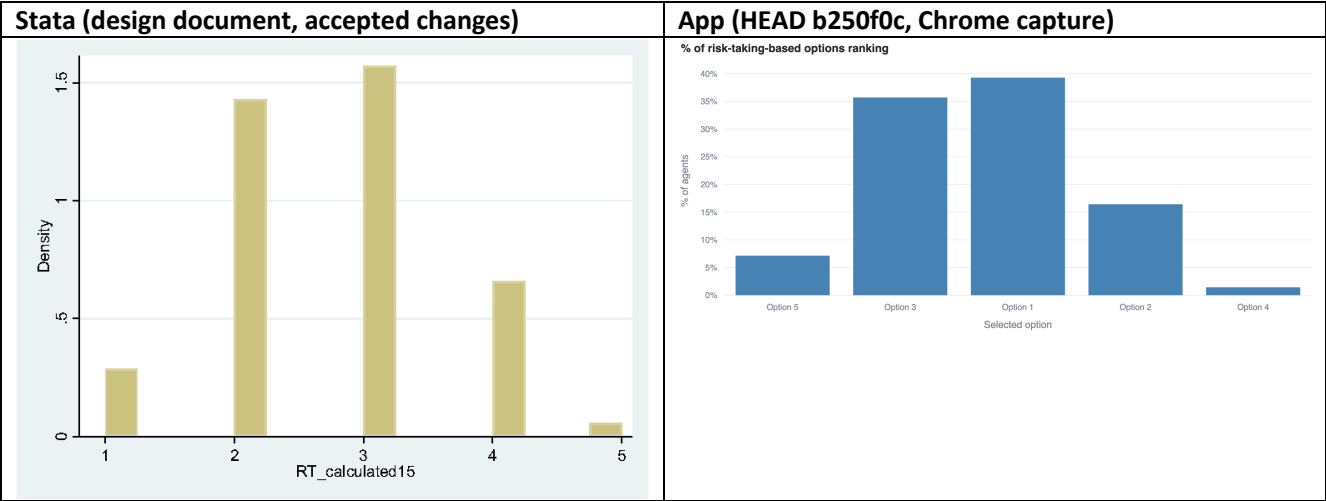
```
gen RT_calculated15 = floor(1+(5-0.0001)*((z_RT_calculated_hs -
min_RT_calculated)/(max_RT_calculated - min_RT_calculated)))
histogram RT_calculated15
```

| Variable        | Stata<br>Obs | Stata<br>Mean | Stata SD  | Stata<br>Min | Stata<br>Max | App<br>Obs | App<br>Mean | App SD    | App<br>Min | App<br>Max |
|-----------------|--------------|---------------|-----------|--------------|--------------|------------|-------------|-----------|------------|------------|
| RT_calculated15 | 280          | 2.692857      | 0.8790385 | 1            | 5            | 280        | 2.692857    | 0.8790385 | 1          | 5          |

| Category  | Stata count | Stata % | App count | App % | Equal |
|-----------|-------------|---------|-----------|-------|-------|
| segment 1 | 20          | 7.1%    | 20        | 7.1%  | yes   |

|           |     |       |     |       |     |
|-----------|-----|-------|-----|-------|-----|
| segment 2 | 100 | 35.7% | 100 | 35.7% | yes |
| segment 3 | 110 | 39.3% | 110 | 39.3% | yes |
| segment 4 | 46  | 16.4% | 46  | 16.4% | yes |
| segment 5 | 4   | 1.4%  | 4   | 1.4%  | yes |

| Category       | Stata count | Stata % | App count | App % | Equal |
|----------------|-------------|---------|-----------|-------|-------|
| first option 1 | 110         | 39.3%   | 110       | 39.3% | yes   |
| first option 2 | 46          | 16.4%   | 46        | 16.4% | yes   |
| first option 3 | 100         | 35.7%   | 100       | 35.7% | yes   |
| first option 4 | 4           | 1.4%    | 4         | 1.4%  | yes   |
| first option 5 | 20          | 7.1%    | 20        | 7.1%  | yes   |



The values match exactly: z\_RT\_calculated\_hs 280/280 (max |diff| 1.7e-07), RT\_calculated15 280/280, choice lists 280/280.

Step 8: Risk-Taking ranking, categorical income – T2, F10, F11

Settings as in step 6 (β3 = 0, Categorical only).

```
regress RT_calculated_hs RT_noincome_hs i.totalallowance
    32 .0026128 | 72 .0050555 | 128 .0092738 | 200 .0179812 | _cons -.0068307
gen weighted_RT_categorical = cond(assignedallowancelevel ==5, 0.0179812-0.0068307 + ...,
cond(... ==4, 0.0092738-0.0068307 + ..., cond(... ==3, 0.0050555-0.0068307 + ..., cond(... ==2,
0.0026128-0.0068307 + ..., -0.0068307 + ...)))
egen z_RT_categorical = std(weighted_RT_categorical)
replace z_RT_categorical = z_RT_categorical+ beta3
```

| Quintile  | Stata (regress, document)           | App β_income_q | Equal |
|-----------|-------------------------------------|----------------|-------|
| Q1 (€12)  | -.0068307                           | -0.0068307     | yes   |
| Q2 (€32)  | -.0068307 + .0026128 =<br>-.0042179 | -0.0042179     | yes   |
| Q3 (€72)  | -.0068307 + .0050555 =<br>-.0017752 | -0.0017752     | yes   |
| Q4 (€128) | -.0068307 + .0092738 =<br>.0024431  | 0.0024431      | yes   |
| Q5 (€200) | -.0068307 + .0179812 =<br>.0111505  | 0.0111505      | yes   |

Deploy ⓘ

Sub-Decision Mechanisms

1. Options List Length (Tendency to Plan) 2. Loyalty Ranking 3. Willingness-to-Pay Ranking 4. Risk-Taking Ranking 5. Flexibility Ranking 6. Integrated Default List

Estimates Risk-Taking propensity based on Big-5 personality traits and income.

$RiskTaking_i = \beta_3 + 0.0259423863 \times z_{Extroversion_i} + 0.0236992149 \times z_{Openness_i} - 0.0387343152 \times z_{Agreeableness_i} - 0.0377394407 \times z_{Conscientiousness_i} - 0.0253886979 \times z_{Neuroticism_i}$

Income Quintile Effects (β\_income\_q):

| Quintile  | β_income_q |
|-----------|------------|
| Q1 (€12)  | -0.0068307 |
| Q2 (€32)  | -0.0042179 |
| Q3 (€72)  | -0.0017752 |
| Q4 (€128) | 0.0024431  |
| Q5 (€200) | 0.0111505  |

β\_income\_q: Income quintile effects based on agent's income category (Quintiles 1-5)

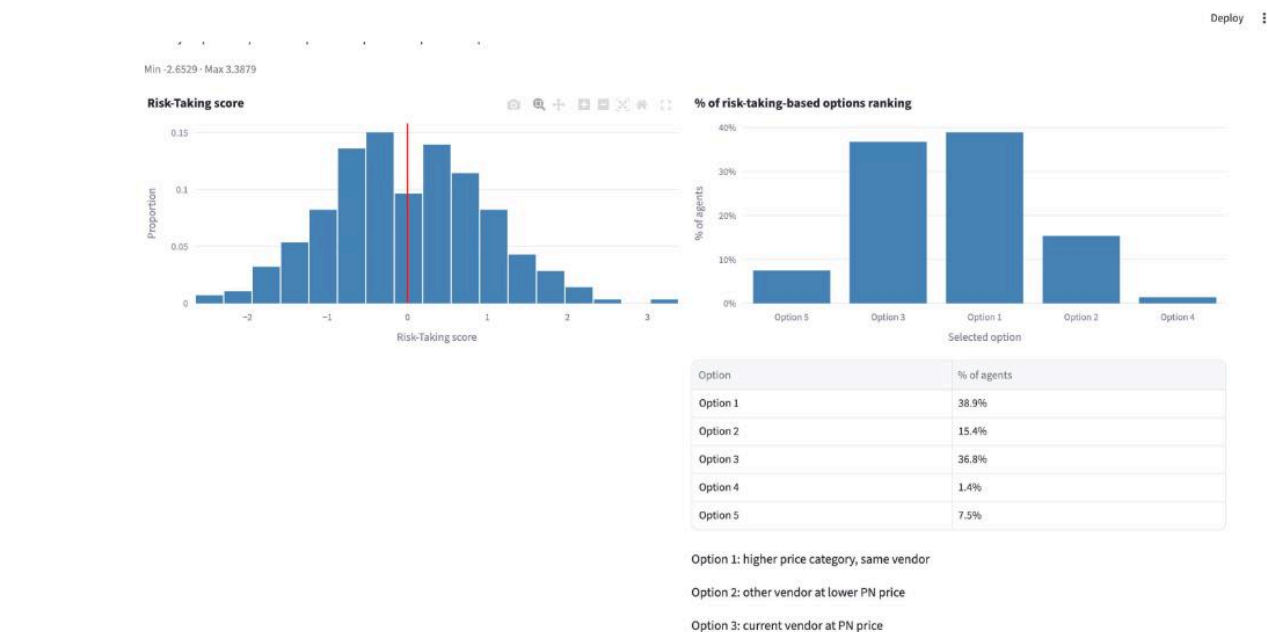
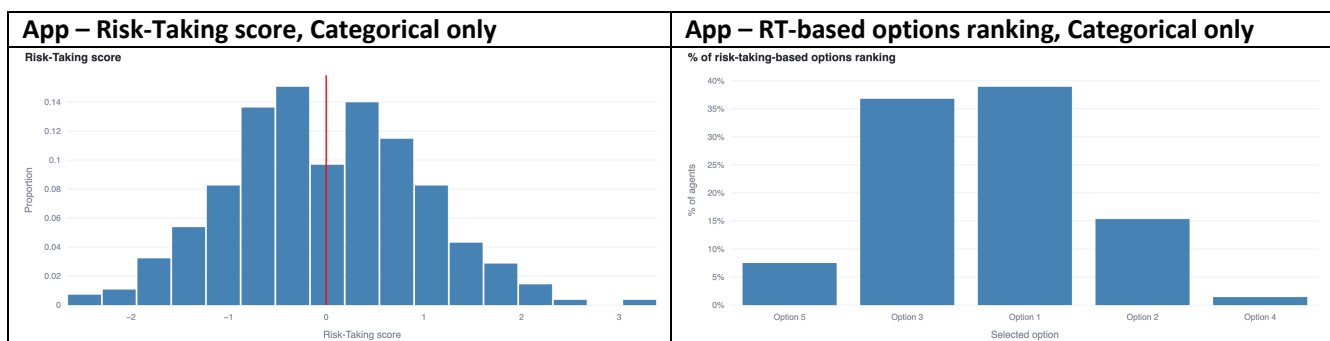
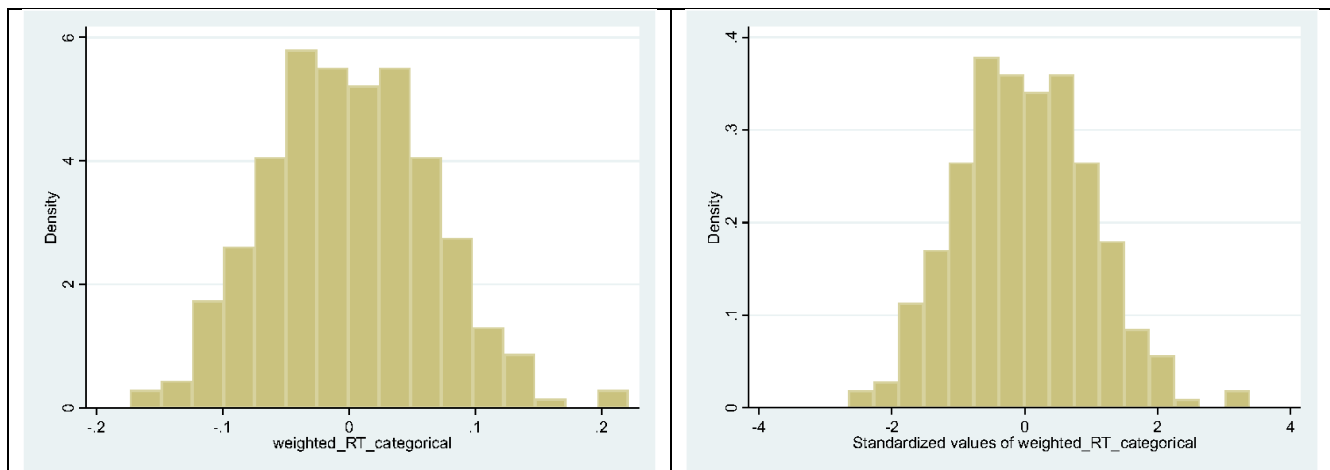
Each value = base intercept (-0.0068307) + the quintile's differential income effect (Quintile 1 is the base level).

$$RiskTaking15_i = \left\lceil 1 + (5 - 0.0001) \times \frac{RiskTaking_i - \min(RiskTaking)}{\max(RiskTaking) - \min(RiskTaking)} \right\rceil \in \{1, \dots, 5\}$$

Risk-Taking rejected transaction options sequence: Option 4 > Option 2 > Option 1 > Option 3 > Option 5

| Risk-Taking score segment                       | Priority list for rejected transaction options | Options list length |
|-------------------------------------------------|------------------------------------------------|---------------------|
| 1 (Lowest 20% of Risk-Taking score<br>cutoff=1) | 5                                              | 1                   |

|                                         |                              |
|-----------------------------------------|------------------------------|
| Stata F10 – raw weighted_RT_categorical | Stata F11 – z_RT_categorical |
|-----------------------------------------|------------------------------|



| Variable         | Stat<br>a<br>Obs | Stata Mean  | Stata SD | Stata Min | Stata Max | Ap<br>p<br>Obs | App Mean     | App SD | App Min   | App Max  |
|------------------|------------------|-------------|----------|-----------|-----------|----------------|--------------|--------|-----------|----------|
| z_RT_categorical | 280              | 8.53158e-10 | 1        | -2.652918 | 3.387871  | 280            | 9.516197e-18 | 1      | -2.652918 | 3.387871 |

|                               |     |               |            |            |           |     |               |            |            |           |
|-------------------------------|-----|---------------|------------|------------|-----------|-----|---------------|------------|------------|-----------|
| weighted_RT_categorical (raw) | 280 | -4.175324e-08 | 0.06530878 | -0.1732589 | 0.2212577 | 280 | -5.230291e-08 | 0.06530878 | -0.1732589 | 0.2212577 |
|-------------------------------|-----|---------------|------------|------------|-----------|-----|---------------|------------|------------|-----------|

The document stores no categorical RT segments; applying its floor rule to the .dta's z\_RT\_categorical gives 21/103/109/43/4, identical to the app's categorical RT segments.

| Category                              | Stata count | Stata % | App count | App % | Equal |
|---------------------------------------|-------------|---------|-----------|-------|-------|
| segment (floor rule on .dta vs app) 1 | 21          | 7.5%    | 21        | 7.5%  | yes   |
| segment (floor rule on .dta vs app) 2 | 103         | 36.8%   | 103       | 36.8% | yes   |
| segment (floor rule on .dta vs app) 3 | 109         | 38.9%   | 109       | 38.9% | yes   |
| segment (floor rule on .dta vs app) 4 | 43          | 15.4%   | 43        | 15.4% | yes   |
| segment (floor rule on .dta vs app) 5 | 4           | 1.4%    | 4         | 1.4%  | yes   |

The values match exactly: z\_RT\_categorical 280/280 (max |diff| 2.0e-07); floor-rule segments 280/280.

## Step 9: Flexibility ranking – F13, F14, F15, F16

Settings as in step 3 ( $\beta_4 = 0$ , W\_OFlex = 0.25; no income term).

```

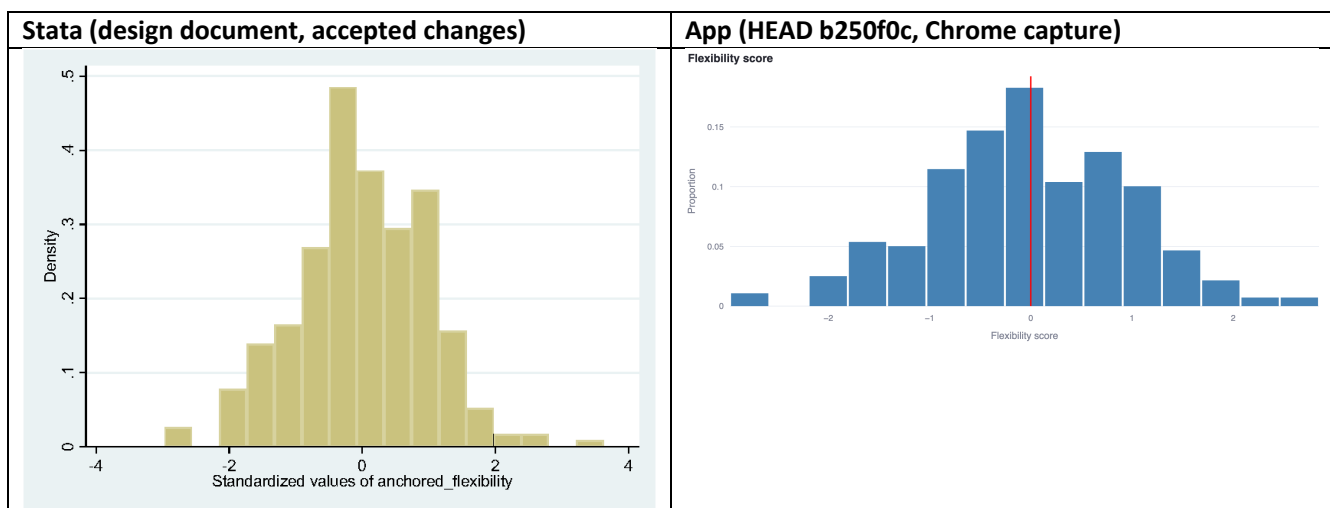
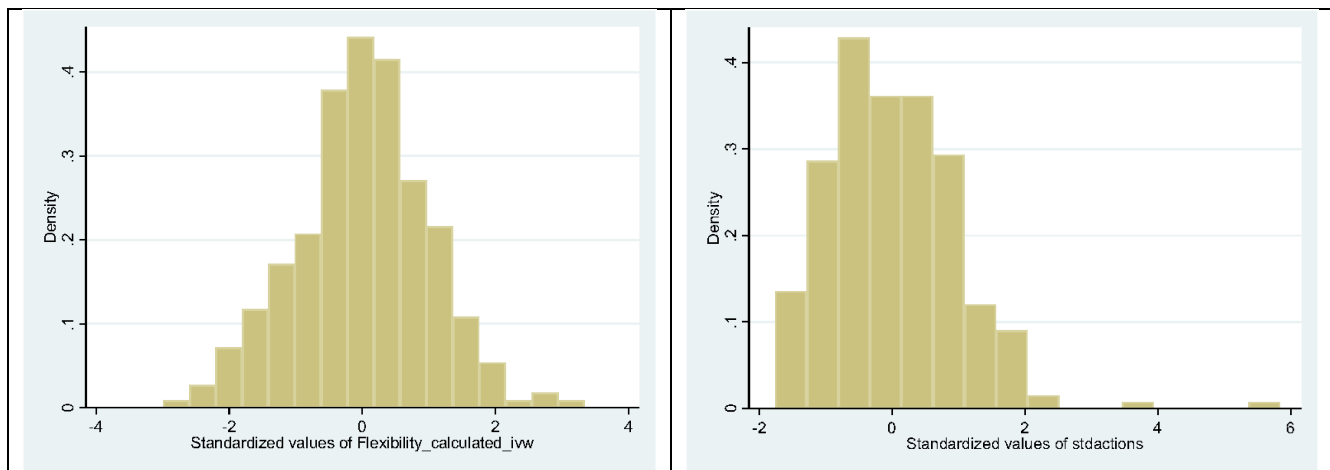
gen Flexibility_calculated_ivw = 0.0206 * z_extraversionbig5 + 0.0294118 * z_opennessbig5 -
0.04921357 * z_neuroticismbig5 + 0.04339814 * z_agreeable + 0.04811179 * z_conscientiousnessbig5
egen z_Flexibility_calculated_ivw = std(Flexibility_calculated_ivw)
replace z_Flexibility_calculated_ivw = z_Flexibility_calculated_ivw + beta4
histogram z_Flexibility_calculated_ivw
egen z_stdactions = std(stdactions)
histogram z_stdactions
gen anchored_flexibility = 0.25*z_stdactions + 0.75 * z_Flexibility_calculated_ivw
egen z_anchored_flexibility = std(anchored_flexibility)
histogram z_anchored_flexibility

```

F13 and F14 are intermediate variables that the app does not chart; they are compared through the app's export columns (rtd\_flex\_z\_ivw, rtd\_z\_stdactions). F15 is the app's "Flexibility score" chart.

| Variable                           | Stat a Obs | Stata Mean   | Stat a SD | Stata Min | Stata Max | App Obs | App Mean      | App SD | App Min   | App Max  |
|------------------------------------|------------|--------------|-----------|-----------|-----------|---------|---------------|--------|-----------|----------|
| z_Flexibility_calculated_ivw       | 280        | 8.764411e-10 | 1         | -2.994517 | 3.338073  | 280     | 1.268826e-17  | 1      | -2.994517 | 3.338074 |
| z_stdactions (.dta: z_stdactionsP) | 280        | 5.554674e-09 | 1         | -1.756797 | 5.835501  | 280     | 4.165653e-08  | 1      | -1.756797 | 5.835501 |
| z_anchored_flexibility             | 280        | 3.891598e-10 | 1         | -2.970918 | 3.623505  | 280     | -2.854859e-17 | 1      | -2.970918 | 3.623505 |

|                                                                         |                                                           |
|-------------------------------------------------------------------------|-----------------------------------------------------------|
| Stata F13 – z_Flexibility_calculated_ivw (export column rtd_flex_z_ivw) | Stata F14 – z_stdactions (export column rtd_z_stdactions) |
|-------------------------------------------------------------------------|-----------------------------------------------------------|

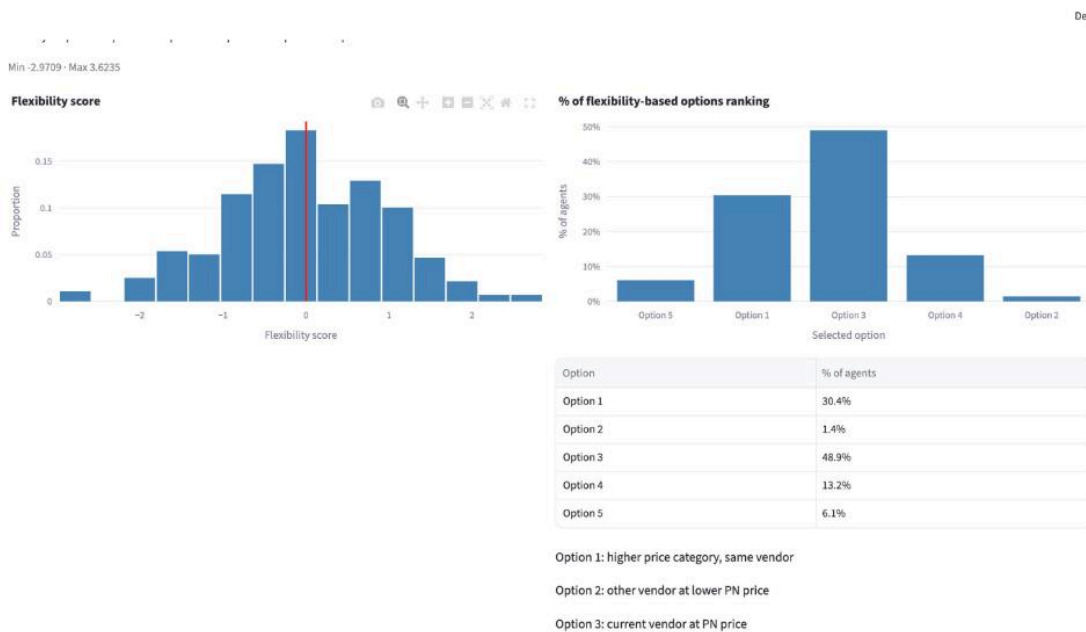
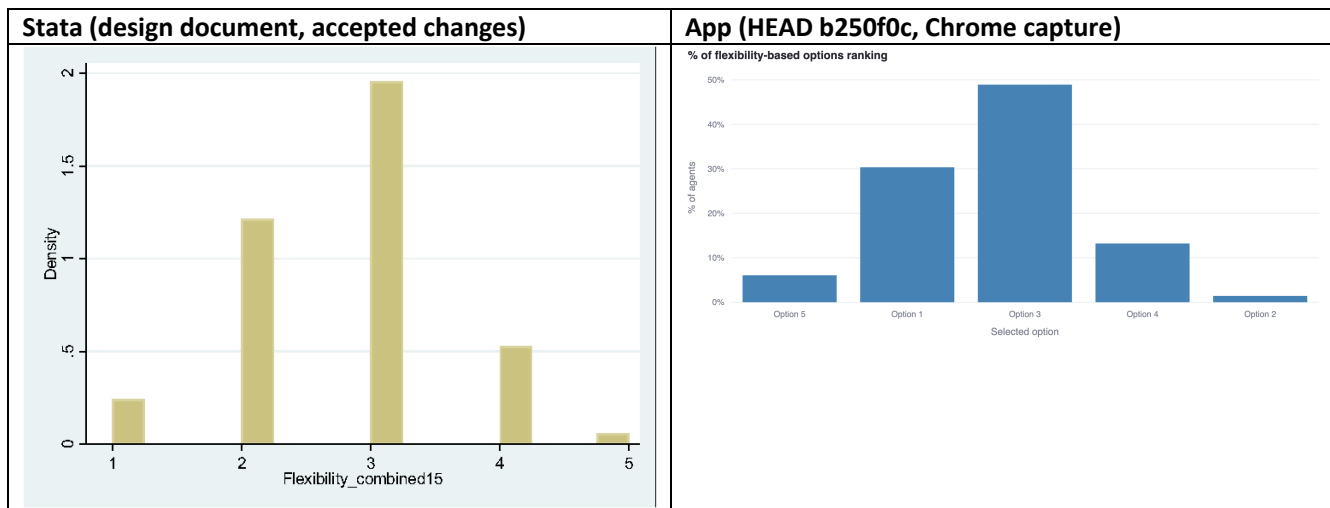


```
gen Flexibility_combined15 = floor(1+(5-0.0001)*((z_anchored_flexibility-
min_Flexibility_combined)/(max_Flexibility_combined - min_Flexibility_combined)))
histogram Flexibility_combined15
```

| Variable               | Stata Obs | Stata Mean | Stata SD | Stata Min | Stata Max | App Obs | App Mean | App SD   | App Min | App Max |
|------------------------|-----------|------------|----------|-----------|-----------|---------|----------|----------|---------|---------|
| Flexibility_combined15 | 280       | 2.735714   | 0.817468 | 1         | 5         | 280     | 2.735714 | 0.817468 | 1       | 5       |

| Category  | Stata count | Stata % | App count | App % | Equal |
|-----------|-------------|---------|-----------|-------|-------|
| segment 1 | 17          | 6.1%    | 17        | 6.1%  | yes   |
| segment 2 | 85          | 30.4%   | 85        | 30.4% | yes   |
| segment 3 | 137         | 48.9%   | 137       | 48.9% | yes   |
| segment 4 | 37          | 13.2%   | 37        | 13.2% | yes   |
| segment 5 | 4           | 1.4%    | 4         | 1.4%  | yes   |

| Category       | Stata count | Stata % | App count | App % | Equal |
|----------------|-------------|---------|-----------|-------|-------|
| first option 1 | 85          | 30.4%   | 85        | 30.4% | yes   |
| first option 2 | 4           | 1.4%    | 4         | 1.4%  | yes   |
| first option 3 | 137         | 48.9%   | 137       | 48.9% | yes   |
| first option 4 | 37          | 13.2%   | 37        | 13.2% | yes   |
| first option 5 | 17          | 6.1%    | 17        | 6.1%  | yes   |



The values match exactly:  $z\_Flexibility\_calculated\_ivw$ ,  $z\_stdactions$  and  $z\_anchored\_flexibility$  280/280 (max  $|diff| \leq 2.2e-07$ ),  $Flexibility\_combined15$  280/280, choice lists 280/280.

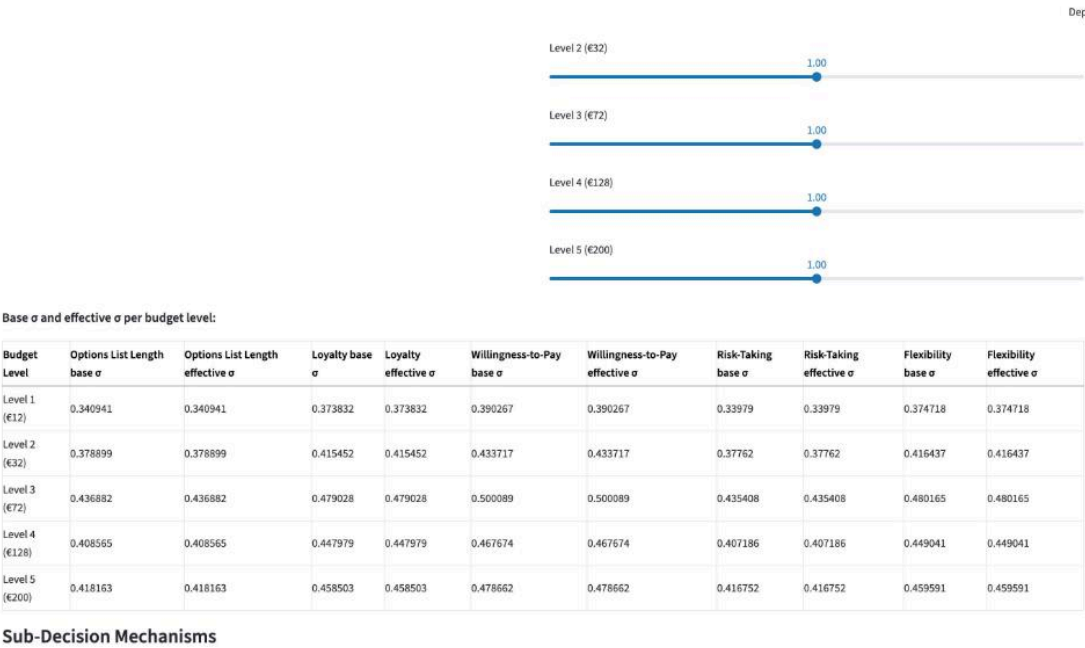
## Step 10: The $\sigma$ tables (stochastic component constants)

The document derives, per element,  $\sigma\_overall = (range/18) \cdot mean(stdactions)$  and  $\sigma\_quintile(q) = (range/18) \cdot mean(stdactions \text{ within allowance group } q)$ .  $mean(stdactions)$  by total allowance from the .dta:

| Total allowance | mean(stdactions) (.dta) |
|-----------------|-------------------------|
| 12              | 1.022823                |
| 32              | 1.136698                |
| 72              | 1.310647                |
| 128             | 1.225694                |
| 200             | 1.25449                 |
| Total           | 1.186338                |

Ranges of the standardised scores (document vs .dta): weighted\_loyalty 7.0783045 vs 7.078304; z\_WTP\_calculated 6.868056 vs 6.868057; z\_RT\_calculated\_hs 5.979747 vs 5.979747; z\_anchored\_flexibility 6.5944228 vs 6.594423 (float32 storage explains the 7th-digit differences).

| Element     | Doc $\sigma$ overall | App base $\sigma$ | Doc $\sigma$ quintile 12/32/72/128/200                                   | App $\sigma$ per level 1–5                                |
|-------------|----------------------|-------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|
| TTP         | 0.395446             | 0.395446          | 0.340941 / 0.378899 / 0.436882 / 0.408565 / 0.418163                     | 0.3409412 / 0.3788994 / 0.4368824 / 0.4085648 / 0.4181634 |
| Loyalty     | 0.4665145336         | 0.4665145         | 0.3738315395 / 0.4154515424 / 0.4790281978 / 0.4479787442 / 0.4585033149 | 0.3738315 / 0.4154515 / 0.4790282 / 0.4479787 / 0.4585033 |
| WTP         | 0.4526575455         | 0.4526575         | 0.3902671709 / 0.4337170115 / 0.5000888363 / 0.4676742829 / 0.4786615699 | 0.3902672 / 0.433717 / 0.5000888 / 0.4676743 / 0.4786616  |
| Risk-Taking | 0.39522204           | 0.395222          | 0.33979032 / 0.37762039 / 0.43540774 / 0.40718566 / 0.41675186           | 0.3397903 / 0.3776204 / 0.4354077 / 0.4071857 / 0.4167519 |
| Flexibility | 0.4346228732         | 0.4346229         | 0.3747183671 / 0.4164371037 / 0.4801645799 / 0.4490414678 / 0.4595910065 | 0.3747184 / 0.4164371 / 0.4801646 / 0.4490415 / 0.459591  |



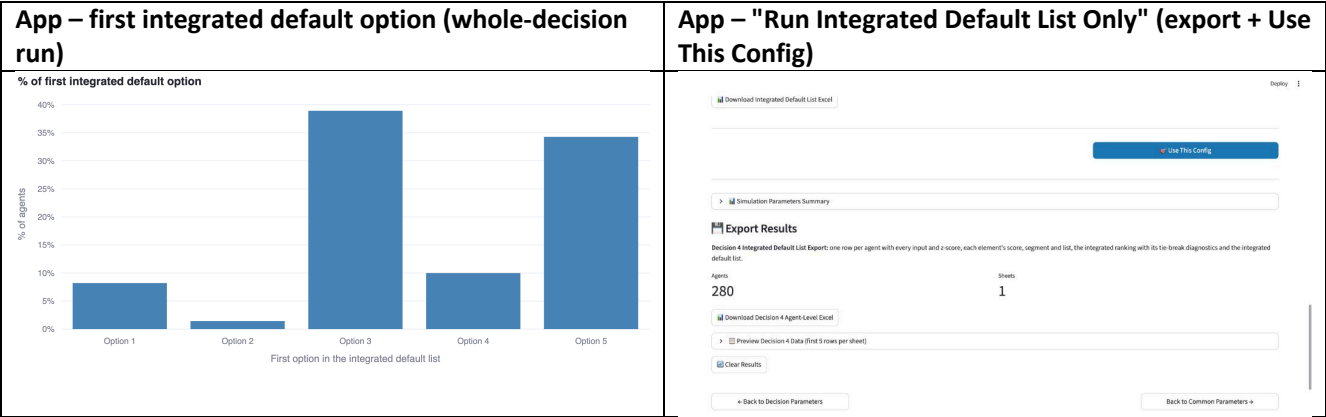
Two arithmetic slips inside the document itself (the app follows the document literally, as instructed): (a) Risk-Taking  $\sigma$  overall is printed as " $0.332208167 \times 1.1863376 = 0.39522204$ ", but that product is 0.39411104 – the per-level column IS consistent with the factor; (b) the Loyalty  $\sigma$  quintile column (0.3738315395 ...) equals the previous revision's column multiplied once more by 1.0228235 and does not equal  $0.393239139 \times \text{mean}(\text{stdactions within$

group) (which would give 0.4022 / 0.4470 / 0.5154 / 0.4820 / 0.4933). TTP, WTP and Flexibility are internally consistent.

The app constants equal the document's tables (every digit); the two inconsistencies above are in the document, not in the app.

Step 11: Integrated default list (document section 6) – app only

The document specifies Kemeny–Young aggregation of the four ranking lists with the tie-breaking hierarchy Schulze → Copeland → footrule → random, truncated to the options list length and at Option 5. It provides no Stata output for it (its stage shares – Kemeny alone 7.7 %, Copeland 43.3 %, footrule 22.7 %, last resort 26.3 % – come from 100 000 random test cases, not from the 280 participants).



| Statistic (Research Baseline, 280, continuous) | App value                                                                         |
|------------------------------------------------|-----------------------------------------------------------------------------------|
| Agents with initial ties after Kemeny          | 98.9% ({'multiple': 215, 'unique_with_ties': 62, 'unique': 3})                    |
| Stage that settled the ranking                 | footrule: 101 (36.1%), random: 93 (33.2%), copeland: 83 (29.6%), kemeny: 3 (1.1%) |
| Final ranking Kemeny-optimal                   | 100.0%                                                                            |
| Truncation rule that bound                     | {'option5': 104, 'length': 100, 'both': 76}                                       |
| Integrated list length 0/1/2/3                 | {'0': 20, '1': 160, '2': 91, '3': 9} (mean 1.32)                                  |
| First integrated option 1...5 (0 = empty list) | {'0': 20, '1': 23, '2': 4, '3': 109, '4': 28, '5': 96}                            |

App-only step: both output rules hold for 280/280 agents (list length ≤ options list length; nothing after Option 5); the integrated-only run and the whole-decision run produce the same frame.

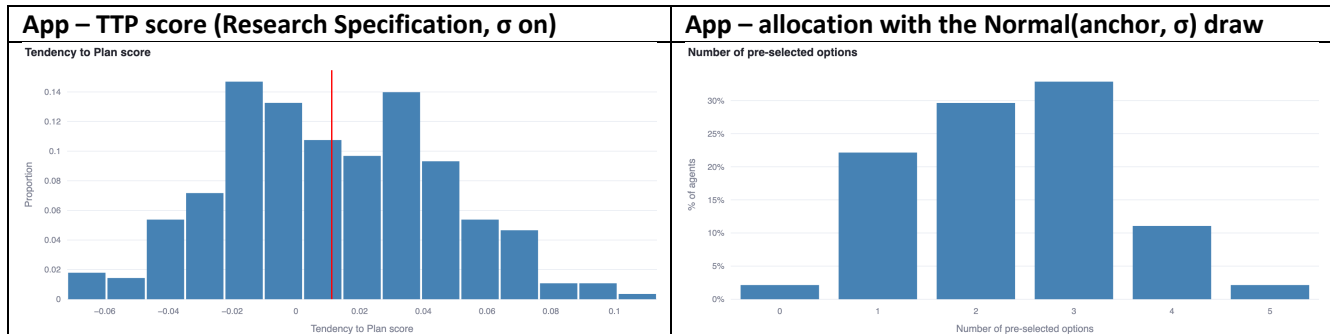
Step 12: Stochastic component – app example (no Stata figure)

Settings: Page 1 Population = Research Specification (same 280 participants, random order), 280 agents, seed 42; Decision 4 tab: "Use Normal(anchor, σ) draw in Research Specification mode" checked, σ uniform, coefficient 1.00 (σ used: TTP 0.395446, Loyalty 0.4665145, WTP 0.4526575, RT 0.395222, Flexibility 0.4346229); intercepts 0.

The score histograms are those of the deterministic scores (identical to steps 3–9); the draws move the allocations. Options List Length allocation, deterministic vs stochastic:

| Options | Deterministic count | % | Stochastic count | % (UI shows 2.1 / 22.1 / 29.6 / 32.9 / 11.1 / 2.1) |
|---------|---------------------|---|------------------|----------------------------------------------------|
|         |                     |   |                  |                                                    |

|   |    |       |    |       |
|---|----|-------|----|-------|
| 0 | 20 | 7.1%  | 6  | 2.1%  |
| 1 | 92 | 32.9% | 62 | 22.1% |
| 2 | 95 | 33.9% | 83 | 29.6% |
| 3 | 57 | 20.4% | 92 | 32.9% |
| 4 | 14 | 5.0%  | 31 | 11.1% |
| 5 | 2  | 0.7%  | 6  | 2.1%  |



App-only step: the on-screen stochastic allocation equals the engine's for the same seed (UI = engine), and the draws are reproducible by seed.

## Step 13: Comparison layouts and Copula

"Compare both" (Decision 4 tab) renders the income-independent elements once and WTP / Risk-Taking / integrated list per income treatment; "Compare all" (Page 1) renders one column per population mode. Each column equals the corresponding single-mode run (verified per agent through the seam).

## COOPECON AI Agent Simulation

### Simulation Results

#### Decision Results

##### ✓ 4. Rejected Transaction Defaults (Custom Parameters)

This decision was configured with custom parameters

⊙ Current Settings: Compare both

Research Baseline

#### Simulation Overview (Research Baseline)

| Total Agents | Avg. Options List Length | Min Options List Length | Max Options List Length |
|--------------|--------------------------|-------------------------|-------------------------|
| 280          | 1.85                     | 0                       | 5                       |

#### Income-Independent Elements

Identical for both income specifications; these elements use no income.

Research Baseline

#### Decision Results

##### ✓ 4. Rejected Transaction Defaults (Custom Parameters)

This decision was configured with custom parameters

⊙ Current Settings: Continuous only

⚙ Copula (Synthetic)

📊 Research Specification

📊 Research Baseline

#### Simulation Overview (Copula, Cont) ⓘ

| Total Agents | Avg. Options List ... | Min Options ... | Max Options ... |
|--------------|-----------------------|-----------------|-----------------|
| 280          | 2.61                  | 0               | 5               |

⚙ Copula (Synthetic)

#### Simulation Overview (Research Spec, Cont)

| Total Agents | Avg. Options List ... | Min Options ... | Max Options ... |
|--------------|-----------------------|-----------------|-----------------|
| 280          | 2.35                  | 0               | 5               |

📊 Research Specification

#### Simulation Overview (Research Baseline, Cont)

| Total Agents | Avg. Options List ... | Min Options ... | Max Options ... |
|--------------|-----------------------|-----------------|-----------------|
| 280          | 1.85                  | 0               | 5               |

📊 Research Baseline

#### Options List Length (Tendency to Plan)

presents how many default options each agent pre-selects (0-5)

Min -0.0952 · Max 0.1101

Tendency to Plan score



#### Options List Length (Tendency to Plan)

presents how many default options each agent pre-selects (0-5)

Min -0.0718 · Max 0.1380

Tendency to Plan score



#### Options List Length (Tendency to Plan)

presents how many default options each agent pre-selects (0-5)

Min -0.0718 · Max 0.1380

Tendency to Plan score



As in the Decision 1 comparison, the Copula column cannot match the Stata values: it is a synthetic population bootstrapped from the fitted copula, never the original 280 participants (here Min  $-0.0952$  / Max  $0.1101$  for the TTP score vs  $-0.0718$  /  $0.1380$ ). Research Specification and Research Baseline carry the original participants; with  $\sigma$  off they give the same multiset of per-agent values (Research Specification permutes the order).

**Consistent: the comparison grids reproduce the single-mode values; Copula differs by construction.**

## Appendix A: Per-participant verification against the .dta (controller script, seam path)

Research Baseline, 280 agents, seed 42, intercepts 0,  $W_{\text{OFlex}}$  0.25,  $\sigma$  off; rows joined on Participant ID.

| Stata column | App column | Participants matching | max  diff |
|--------------|------------|-----------------------|-----------|
|--------------|------------|-----------------------|-----------|

|                                        |                               |                                                                               |         |
|----------------------------------------|-------------------------------|-------------------------------------------------------------------------------|---------|
| weighted_ttp                           | rtd_weighted_ttp              | 280/280                                                                       | 1.8e-08 |
| choice_length_deterministic            | rtd_choice_length             | 280/280                                                                       | 0       |
| weighted_loyalty                       | rtd_loyalty_z                 | 280/280                                                                       | 3.2e-07 |
| weighted_loyalty15                     | rtd_loyalty_segment           | 280/280                                                                       | 0       |
| choice1...5_loyalty_deterministic      | rtd_loyalty_ranking           | 280/280                                                                       | 0       |
| z_WTP_calculated (continuous)          | rtd_wtp_z                     | 280/280                                                                       | 8.1e-06 |
| WTP_calculated15                       | rtd_wtp_segment               | 280/280                                                                       | 0       |
| choice1...5_WTP_deterministic          | rtd_wtp_ranking               | 280/280                                                                       | 0       |
| z_WTP_categorical (stored .dta column) | rtd_wtp_z (categorical)       | 0/280 – superseded formula in the .dta (step 6); app = document code to 9e-09 | 6.2e-02 |
| floor rule on z_WTP_categorical (.dta) | rtd_wtp_segment (categorical) | 279/280                                                                       | –       |
| z_RT_calculated_hs (continuous)        | rtd_rt_z                      | 280/280                                                                       | 1.7e-07 |
| RT_calculated15                        | rtd_rt_segment                | 280/280                                                                       | 0       |
| choice1...5_RT_deterministic           | rtd_rt_ranking                | 280/280                                                                       | 0       |
| z_RT_categorical                       | rtd_rt_z (categorical)        | 280/280                                                                       | 2.0e-07 |
| floor rule on z_RT_categorical         | rtd_rt_segment (categorical)  | 280/280                                                                       | –       |
| z_Flexibility_calculated_ivw           | rtd_flex_z_ivw                | 280/280                                                                       | 1.8e-07 |
| z_stdactionsP                          | rtd_z_stdactions              | 280/280                                                                       | 1.6e-07 |
| z_anchored_flexibility                 | rtd_flex_z                    | 280/280                                                                       | 2.2e-07 |
| Flexibility_combined15                 | rtd_flex_segment              | 280/280                                                                       | 0       |
| choice1...5_flex_deterministic         | rtd_flex_ranking              | 280/280                                                                       | 0       |

## Appendix B: Document / data errors found

- The stored column weighted\_WTP\_categorical in Stata\_File\_Decision4\_050926.dta (and therefore z\_WTP\_categorical and figure F7) was produced with a superseded formula (see step 6); the document's printed cond() code is what the app implements.
- Risk-Taking  $\sigma_{\text{overall}}$ : the document prints  $0.332208167 \times 1.1863376 = 0.39522204$ ; the product is 0.39411104.
- Loyalty  $\sigma_{\text{quintile}}$  column: the document's five per-allowance values are the previous revision's column multiplied by 1.0228235 again and do not follow from the stated factor 0.393239139.
- Flexibility: the document lists "histogram z\_Flexibility\_calculated\_hs" and Risk-Taking "histogram z\_RT\_calculated\_ivw" without the corresponding figures (the IVW/H&S alternatives that were not retained).
- The document still contains tracked changes (deleted previous-revision figures and choice mappings); this comparison uses the accepted-changes text and figures.